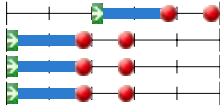
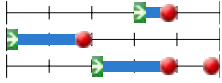
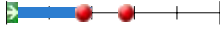
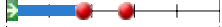
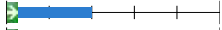
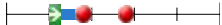


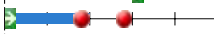
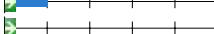
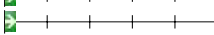
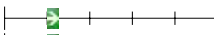
Program overview

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Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master AM 2022							
Common Core AM 2022							
Common Core AM 2022							
TPM005A	Scientific Writing for Applied Mathematics	3					
WM1028AM	Ethics for Applied Mathematics	3					
Orientation Courses AM 2022							
WI4201	Scientific Computing	6					
WI4227-14	Discrete Optimisation	6					
WI4430	Martingales, Brownian Motion	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4645	Introduction to Quantum Information and Computing	6					
WI4655	Perturbation and Variational Methods for Partial Differential Equations	6					
WI4675	Introduction to Financial Mathematics	6					
Specialisation Computational Science and Engineering AM 2022							
Compulsory Courses (select 3 out of 4)				Compulsory Courses Computational Science and Engineering AM 2021 AM 2022			
WI4019-SP	Non-linear Differential Equations	6					
WI4204	Advanced Modeling	6					
WI4205	Applied Finite Elements	6					
WI4475	Data Assimilation	6					
Recommended Courses				Recommended Courses Computational Science and Engineering AM 2022			
IN4049TU	Introduction to High Performance Computing	6					
WI4011-17	Computational Fluid Dynamics	6					
WI4207	Continuous Optimization	6					
WI4209	Systems and Control	6					
WI4212	Advanced Numerical Methods	6					
WI4221	Control of Discrete-Time Stochastic Systems	6					
WI4226	Advanced System Theory	6					
WI4260TU	Scientific Programming for Engineers	3					
WI4450	Special Topics in Computational Science and Engineering	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4655	Perturbation and Variational Methods for Partial Differential Equations	6					
WI4771TU	Object Oriented Scientific Programming with C++	3					
Specialisation Financial Engineering AM 2022							
Compulsory Courses				Compulsory Courses for Financial Engineering AM 2022			
WI4079	Financial Mathematics	6					
WI4154	Computational Finance	6					
WI4201	Scientific Computing	6					
WI4430	Martingales, Brownian Motion	6					
Recommended Courses				Recommended Courses for Financial Engineering AM 2022			
WI4019-SP	Non-linear Differential Equations	6					
WI4052	Risk Analysis	6					
WI4156(TU)	Game theory	6					
WI4205	Applied Finite Elements	6					
WI4224	Special topics in Financial Engineering	6					
WI4230	Time Series & Extreme Value Theory	6					
WI4425	Financial Markets Theory	6					
WI4505	Quantitative Risk Management	6					
WI4614	Stochastic Simulation	6					
WI4630	Statistical Learning	6					
WI4635	Linear Algebra and Optimization for Machine Learning	6					
WI4771TU	Object Oriented Scientific Programming with C++	3					
Specialisation Discrete Mathematics and Optimization AM 2022							
Compulsory Courses				Compulsory Course for Discrete Mathematics and Optimization AM 2022			
WI4620	Semidefinite Optimization	6					
Compulsory Courses (select 2 out of 4)				Compulsory Courses for Discrete Mathematics and Optimization(select 2 out of 4) AM 2022			

WI4410	Advanced Discrete Optimization	6		
WI4515	Relaxations and Heuristics	6		
WI4635	Linear Algebra and Optimization for Machine Learning	6		
WI4670	Extremal Combinatorics	6		
Recommended Courses		Recommended Courses for Discrete Mathematics and Optimization AM 2022		
ET4030	Error Correcting Codes	4		
WI4207	Continuous Optimization	6		
WI4630	Statistical Learning	6		
Specialisation Stochastics AM 2022				
Compulsory Courses		Compulsory Course for Stochastics AM 2022		
WI4455	Statistical Inference	6		
Compulsory Courses (select 2 out of 6)		Compulsory Courses for Stochastics AM 2022 (select 2 out of 6)		
WI4052	Risk Analysis	6		
WI4230	Time Series & Extreme Value Theory	6		
WI4465	Advanced Topics in Probability	6		
WI4615	Stochastic Calculus	6		
WI4665	Advanced Topics in Statistics	6		
Recommended Courses		Recommended Courses for Stochastics AM 2022		
WI4050	Uncertainty and Sensitivity Analysis	6		
WI4138	Decision Theory/Expert Judgment	6		
WI4156(TU)	Game theory	6		
WI4505	Quantitative Risk Management	6		
WI4614	Stochastic Simulation	6		
WI4630	Statistical Learning	6		
WI4640	High-dimensional probability	6		
Specialisation Mathematics of Data Science AM 2022		Specialisation courses for Mathematics of Data Science AM 2022		
Compulsory Courses		Compulsory Courses for Mathematics of Data Science AM 2022		
WI4630	Statistical Learning	6		
Compulsory Courses (select 2 out of 4)		Compulsory Courses for Mathematics of Data Science AM 2022 (select 2 out of 4)		
WI4203	Applied Functional Analysis	6		
WI4455	Statistical Inference	6		
WI4614	Stochastic Simulation	6		
WI4640	High-dimensional probability	6		
Recommended Courses		Recommended Courses for Mathematics of Data Science AM 2022		
CS4230	Machine Learning 2	5		
WI4207	Continuous Optimization	6		
WI4230	Time Series & Extreme Value Theory	6		
WI4410	Advanced Discrete Optimization	6		
WI4620	Semidefinite Optimization	6		
WI4645	Introduction to Quantum Information and Computing	6		
WI4665	Advanced Topics in Statistics	6		
Specialisation Partial Differential Equations AM 2022		Specialisation courses for Partial Differential Equations AM 2022		
Compulsory Courses (select 3 out of 6)		Compulsory Courses for Partial Differential Equations AM 2022 (select 3 out of 6)		
WI4019-SP	Non-linear Differential Equations	6		
WI4201	Scientific Computing	6		
WI4203	Applied Functional Analysis	6		
WI4430	Martingales, Brownian Motion	6		
WI4520	Nonlinear Analysis and Partial Differential Equations	6		
WI4660	Dynamical Systems and Chaos	6		
Recommended Courses		Recommended Courses for Partial Differential Equations AM 2022		
WI4006	Special Functions and Representation Theory	6		
WI4046	Spectral Theory of Linear Operators	6		
WI4204	Advanced Modeling	6		
WI4205	Applied Finite Elements	6		
WI4211	Advanced Topics in Analysis	6		
WI4480	Internet Seminar on Evolution Equations	9		
WI4485	Harmonic Analysis	6		
WI4615	Stochastic Calculus	6		
Specialisation Mathematics of Quantum Technology and Computation AM		Specialisation courses for Mathematics of Quantum Technology and Computation AM 2022		
Compulsory Courses (select 2 out of 4)		Compulsory Courses Mathematics of Quantum Technology and Computation AM 2022 (select 2 out of 4)		
CS4090AM	Quantum Communication and Cryptography	6		

WI4006	Special Functions and Representation Theory	6	
WI4203	Applied Functional Analysis	6	
WI4620	Semidefinite Optimization	6	
Recommended courses			
Recommended courses Mathematics of Quantum Technology and Computation AM 2022			
AP3663	Special Topics in Quantum Technology	4	
WI4046	Spectral Theory of Linear Operators	6	
WI4207	Continuous Optimization	6	
WI4211	Advanced Topics in Analysis	6	
WI4635	Linear Algebra and Optimization for Machine Learning	6	
WI4650	Applied Quantum Algorithms	6	
Non-Mathematics Electives AM 2022			
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
TUD4040	Joint Interdisciplinary Project	15	
WI5012	Internship 12 EC	12	
WI5118	Internship 18 EC	18	
WM0375TU	Ethics of Technological Risks	3	
WM1135TU	Advanced English for the University	3	
WM1301TU	Ethics of Transportation	3	
WM1401TU	Ethics of Healthcare Technologies	3	
Thesis project			
WI5005	Thesis Project	42	
WI5005EXT	Thesis Project (Extern, with additional internship optional)	42	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Master AM 2022

MSc Coordinator	Dr.ir. J.H. Weber
Program Title	Applied Mathematics
In association with the University of	TU/e and UTwente
Introduction 1	The Master of Science programme in Applied Mathematics is intended for students who have obtained a Bachelor of Science degree in (Applied) Mathematics or in another area that provides a sufficiently advanced background in mathematics. Students may apply for admission through the admissions office of the Delft University of Technology. Any student who has completed a Bachelor of Science degree in (Applied) Mathematics at a Dutch research university is automatically qualified for admission to the M.Sc. programme. Also, a completed bridging programme in Applied Mathematics from TU Delft gives access. Individuals holding a foreign bachelor's degree will be assessed on an individual basis by the admission committee. The objective of the MSc programme in Applied Mathematics is to provide students with a thorough knowledge of applied mathematics and to develop their expertise in applying the methods and tools of mathematics to problems in science and engineering. It is therefore crucial for students to obtain broad experience in the simulation and mathematical modeling of such problems, the mathematical analysis of these models and the implementation of the results. Moreover, they should have sufficient insight into the underlying mathematical theory to be able to develop new mathematical methods and techniques, if needed. The M.Sc. programme in Applied Mathematics is a two-year full-time programme, to be concluded with a masters thesis. In total, the programme requires 120 EC.
Program Structure 1	The Master's Degree Programme in Applied Mathematics has seven specialisations: - Computational Science and Engineering, - Financial Engineering, - Discrete Mathematics and Optimization, - Stochastics, - Mathematics of Data Science, - Partial Differential Equations, - Mathematics of Quantum Technology and Computation. An Individual Exam Programme (IEP) consists of (at least) 120 EC, subdivided into the following parts: - Common core courses worth 6 EC, - Orientation course of 6 EC, - Specialisation courses worth 48-54 EC, of which a maximum of 12 EC may be spent on non-mathematical specialisation courses, - Non-mathematical electives worth 12-18 EC, - A thesis project of 42 EC. During their education, students are required to gain experience as a mathematician in interdisciplinary work. This requirement can be satisfied in three ways: - through an internship of 12 or 18 EC as part of the non-mathematics electives; - through a Joint Interdisciplinary Project; - through a graduation project performed in an interdisciplinary setting outside the Delft Institute of Applied Mathematics. Students that have been admitted through a bridging programme (i.e. with a bachelor degree different from (Applied) Mathematics) or that are following the SEC specialisation are exempt from this requirement. The joint masters programme in Computer Simulations for Science and Engineering (COSSE) deviates. Students who have followed the minor Science Education and Communication (SEC) in their bachelors programme and wish to obtain the certificate granting 1st degree teacher (eerstegraads bevoegdheid) status in the Netherlands may follow the SEC specialisation (Verdiepingsdeel) at the faculty of Applied Science within their Applied Mathematics Masters programme.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Common Core AM 2022

Introduction 1	The core consists of 1. The common core courses and 2. The mandatory orientation course for each specialisation
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Year

Organization

Education

2022/2023

Electrical Engineering, Mathematics and Computer Science

Master Applied Mathematics

Common Core AM 2022	
Introduction 1	1. The common core consists of two courses, TPM005A Scientific Writing for Applied Mathematics and WM1028AM Ethics for Applied Mathematics

TPM005A	Scientific Writing for Applied Mathematics	3
Module Manager	Dr. K. van Nispen	
Instructor	Drs. A.E. Kam	
Instructor	M. Looij	
Instructor	A. Glasbergen-Plas	
Instructor	Dr. K. van Nispen	
Instructor	L. Meester	
Instructor	Drs. P.C. Post	
Co-responsible for assignments	Drs. P.C. Post	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	This course is mandatory for all first-year Master students of Applied Mathematics.	
Course Contents	In this course, you learn to write a scientific article, more specifically: a literature review about a subject of your own choice within the field of applied mathematics. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master programme, such as your Master thesis. In seven weeks, we will go through all the steps of the writing process, from formulating a good main question to the actual writing, revising, reviewing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, constructing your argumentation, reviewing fellow students' articles and using the instructor's and other students feedback to improve your work	
Study Goals	At the end of this course, you will be able to write a literature review. To achieve this, at the end of this course: Students are able to identify the relevant author guidelines for their text genre Students are able to formulate a main question (or aim/purpose statement) with subquestions Student support writing sufficiently and correctly with literature Students support writing with argumentation Students can structure a text according to the scientific conventions Students can implement knowledge about the specific elements of a scientific text and their generic requirements in their own text. Students use visualisations, such as tables and figures, appropriately. Students apply a correct formal scientific style	
Education Method	The purpose of this course is to write a literature review about a mathematical subject of your choice, together with a fellow student. You each read at least five relevant sources about this subject and combine these in a meaningful way in your text. Every week you read some background information about scientific writing and hand in writing assignments, following the steps in the writing process. During the face-to-face (online or on-campus) sessions we discuss the writing assignments and you will have the opportunity to ask questions and discuss aspects of scientific writing with the instructor and with your fellow students. To get a grade for this course, you must attend (online or on-campus) face-to-face sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 3 ECTS. This equals 84 hours of study. A total of 12 hours is allotted to attending the (online or on-campus) sessions; the remaining 72 hours is for self study, writing and communication with your fellow writer. Usually you will focus on reading the course materials and background literature in the first weeks of the course. The time allotted to writing and revising will increase as the course proceeds. Make sure to schedule this time in your personal agenda.	
Literature and Study Materials	Theory about academic writing will be made available through Brightspace.	
Assessment	Together with a fellow student you write and submit a literature review of 4000 words, excluding the abstract and the reference list ((10% margin). You have to hand in parts of your article every week and revise parts that you have already written. Your final grade is based on the final article, your attendance at the face-to-face sessions and in-time submission of the assignments during the course. The review will be graded by way of an evaluation form that will be made accessible throughout the course.	
Special Information	For more information about this course, contact the coordinator, Dr. Karin van Nispen (k.vannispen@tudelft.nl)	
Elective	Yes	
Category	MSc level	

WM1028AM	Ethics for Applied Mathematics	3
Module Manager	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course is a study of broadly normative issues that arise in relation to applied mathematics. The following kinds of questions are examined: How can applied systems be built to promote, protect, preserve, and foster the pursuit of various values related to human wellbeing? What risks and uncertainties need to be accounted for in such systems? To what extent is it possible to empower individuals with applied systems while limiting the extent to which they can harm each other and themselves?	
Study Goals	1. Recognize and discuss normative and societal implications of mathematical applications and cases. 2. Explain key course concepts (e.g., bias, normativity, modeling assumptions) and their ethical dimensions. 3. Apply basic ethical frameworks and concepts to mathematical applications in society. 4. Think critically about the roles and responsibilities of mathematics professionals. 5. Listen, understand and respond to people with different viewpoints and ideas. Disagreement and debate is expected, but so is respectful and open communication.	
Education Method	1 lecture and 1 tutorial per week	
Literature and Study Materials	Readings will be provided on Brightspace.	
Assessment	Final assignment: 35% (individual) Tutorial Presentation: 20% (small groups) Tutorial Participation: 20% (individual) Ethical Assessment paper: 25% (small groups)	
Remarks	Contact the course instructor for disability accommodations.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Orientation Courses AM 2022

Introduction 1	2. The second part of the core of your IEP consists of an orientation course, which is compulsory for your specialisation. WI4201 Scientific Computing, mandatory for Computational Science and Engineering WI4227-14 Discrete Optimisation, mandatory for Discrete Mathematics and Optimization WI4430 Martingales, Brownian Motion, mandatory for Stochastics WI4635 Linear Algebra and Optimization for Machine Learning, mandatory for Mathematics of Data Science WI4645 Introduction to Quantum Information and Computing, mandatory for Mathematics of Quantum Technology and Computation WI4655 Perturbation and Variational Methods for PDE, mandatory for Partial Differential Equations WI4675 Introduction to Financial Mathematics, mandatory for Financial Engineering It is allowed to take more than one orientation course, you can include the extra course(s) under your specialisation in your IEP.
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WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5.	
Co-Instructor	disclaimer: information may change depending on the developments around the coronavirus. Prof.dr.ir. M.B. van Gijzen	

WI4227-14	Discrete Optimisation	6
Responsible Instructor	Dr. F.M. de Oliveira Filho	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge: * Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: * Proficiency in reading and writing proofs * Sufficient mathematical maturity	
Course Contents	Discrete optimization is about the problem of finding a best solution among a (finite) set of feasible solutions. A well-known example is the traveling salesman problem, where we are asked to find a shortest tour in a graph visiting every node exactly once. This course is an introduction to the area, focusing on fundamental methods like linear programming and its theoretical framework (polyhedra, integrality, etc.). The following topics will be covered: - Convex sets and their structure. - Polyhedra, cones, and their structure. - Linear programming, the simplex method. - Systems of linear inequalities, Fourier-Motzkin elimination. - Farkas's lemma, linear programming duality. - Several applications and examples: flows in networks, matrix games, bipartite matching, etc.	
Study Goals	By the end of this course, students will be able to - Recognize convex sets and their applications; - Apply the theory of convexity to concrete examples; - State the main results/theorems concerning polyhedra; - Model problems as linear programming problems; - Recognize total unimodularity and use it to prove integrality of polyhedra; - Construct the dual of a linear program.	
Education Method	Lectures	
Literature and Study Materials	The course is based on lecture notes and no specific book will be followed.	
Books	No specific book will be followed in this course. The following books contain most of the course's material: * A. Barvinok, A Course in Convexity, American Mathematical Society, Providence, 2002. * V. Chvatal, Linear Programming, W.H. Freeman and Company, New York, 1983. * W.J. Cook, W.H. Cunningham, W.R. Pulleyblank, and A. Schrijver, Combinatorial Optimization, John Wiley & Sons, New York, 1998. * A. Schrijver, Theory of Linear and Integer Programming, John Wiley & Sons, New York, 1986.	
Prerequisites	* Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: Proficiency in reading and writing proofs Sufficient mathematical maturity	
Assessment	Up to 20% of the final grade will be determined by exercise sheets, the rest will be determined by a final exam.	
Permitted Materials during Tests	No materials are permitted during the test (except pencil, ruler, eraser etc). In particular, calculators are not allowed.	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut "A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut "A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	Two graded homeworks and a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f \geq 5$. disclaimer: information may change depending on the developments around the coronavirus. The homework grade does not count for the resit.	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will deal with machine learning from the perspective of linear algebra and optimization. In particular, we will discuss how different machine learning techniques use classical concepts from linear algebra and optimization and how their performance can be improved by using more sophisticated linear algebra and optimization algorithms. We cover theoretical aspects as well as the implementation and application of these techniques. Among others, we will study + efficient matrix operations such as matrix decompositions and factorizations, + graph-related algebra, + first- and second-order optimization algorithms with their learning-oriented customization and regularization techniques, and + optimization on computational graphs (backpropagation).	
Study Goals	The student has gained an understanding of how linear algebra and optimization algorithms are used in and how these are important for the performance and efficiency of different machine learning techniques: clustering, classification and regression, neural networks, matrix decomposition/factorizations Furthermore, the student is able to + motivate and implement different matrix decomposition methods + analyze and explain matrix factorization methods + analyze graph-based data using linear algebraic tools + implement and customize first- and second- order unconstrained optimization methods + analyze and implement constrained and duality-based optimization algorithms + analyze and efficiently implement forward and backward propagation through neural networks	
Education Method	Lectures	
Assessment	Group assignments/projects and a written exam	
Co-Instructor	Dr. K.S. Postek	

WI4645	Introduction to Quantum Information and Computing	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic probability theory. Linear algebra.	
Course Contents	Quantum mechanics is not a prerequisite. We outline the mathematical framework of quantum mechanics, and give an introduction to quantum information theory, quantum computing and quantum algorithms. Since this is the orientation course for the specialization 'Mathematics of quantum technology and computation', we also pay special attention to the role in quantum theory by some of the other topics covered in this specialization: functional analysis, spectral theory, representation theory, semidefinite optimization and Lie theory.	
Study Goals	- Explain the basic postulates of quantum mechanics in terms of operators on Hilbert spaces. - Analyse self-adjoint operators using spectral theory. - Analyse single particle systems (such as qubits and harmonic oscillators) in this setting, and describe composite systems using tensor products. Analyse states on composite systems in terms of entanglement, and explain the Bell inequalities. - Analyse the time evolution on (mixed) states in the Schroedinger picture as well as in the Heisenberg picture. For qubits, explain how this gives rise to the Bloch sphere. - Analyse operations between different quantum systems in terms of completely positive maps (such as qubit gates and POVMs). Explain the Stinespring theorem and the Choi/Kraus decomposition, and explain basic no-go theorems such as the no-cloning theorem. - Analyse (the performance of) simple quantum algorithms (such as FFT). - Explain the connection between symmetries, conserved quantities and Lie theory, and analyse spin and momentum in terms of rotation and translation symmetry.	
Education Method	-2 hours of lectures per week, 14 weeks in total. -Biweekly exercises.	
Literature and Study Materials	We will use the book by Nielsen and Chuang - Quantum computation and quantum information - 10th edition. In addition, we use lecture notes that will be provided via Brightspace.	
Books	Nielsen and Chuang - Quantum computation and quantum information (10e edition). Lecture notes (provided via Brightspace)	
Assessment	A written final exam.	

WI4655	Perturbation and Variational Methods for Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	The course consists of two part. In the first part of the course (Q1), methods will be introduced to get approximate solution to PDEs. In the lectures we elaborate on asymptotic approximations, matched asymptotic expansions, multiple scales, WKB method, bifurcation and stability analysis. In the second part (Q2), a modern introduction to the field of elliptic PDEs is given. In the lectures we consider weak solutions to elliptic equations (such as Laplace equation), Sobolev spaces. Modern PDEs uses tools from functional analysis and measure theory. The precise regularity and behavior of solutions of elliptic equations is one of the main features in the lectures. For non-linear PDEs such results are fundamental. The course might be useful for student in analysis, PDE and numerics, or anyone who wants to know more about Sobolev spaces and second order equations. (Key words: elliptic, weak derivatives, Sobolev spaces extension theorem, trace theorem, Sobolev inequalities, Hilbert techniques, compactness theorems, second order elliptic equations, Weak solution, energy estimates, regularity, eigenvalue techniques).	
Study Goals	Global objectives for the first part of the course: After following this course the student will be able to apply fundamental and advanced analytical techniques and methods to study properties of solutions of ordinary and partial differential equations. The mathematical methods and techniques that the student is able to use are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations. Global objectives for the second part of the course: After the course the student - knows what a weak derivative is - understands the definition of L^p and Sobolev spaces. - can apply basic Sobolev space techniques (approximation, extension, trace, Sobolev embedding, compactness) - can give the definition of a weak solution of a second order elliptic PDE - can apply Sobolev space techniques to prove existence and uniqueness of weak solutions of second order elliptic PDEs - can derive properties (regularity/estimates) of solutions of second order elliptic PDEs Objective per topic: The student Sobolev spaces the student -knows the definition of Hölder spaces -knows the definition of the weak derivative -can calculate weak derivatives -knows the definition of Sobolev spaces -can approximate Sobolev functions by smooth functions -can explain the extension theorem -can explain the trace theorem -knows Sobolev inequalities -can apply Sobolev inequalities -can derive Sobolev inequalities -knows the compactness theorem -can apply the compactness theorem -understands what the dual space of a Sobolev space is Second order elliptic equations the student -knows the definition of a weak solution. -can give energy estimates -can apply the Lax-Milgram theorem to obtain existence of weak solutions -knows what regularity is -can state regularity results for second order equations (interior and boundary) -can apply eigenvalue techniques.	
Education Method	First part of the course: Introduction to Perturbation Methods (second edition) by M.H. Holmes Second part of the course: lecture notes. Optional literature: Partial Differential Equations, by Lawrence C. Evans, AMS, 1998, 662 pp., Hardcover, ISBN 0-8218-0772-2. Assessment	
Assessment	Two partial exams, one at the end of each period. Minimum of a 5 for each partial exam, the final grade is the average of both grades.	
Co-Instructor	Dr.ir. M.C. Veraar	

WI4675	Introduction to Financial Mathematics	6
Responsible Instructor	Dr. A. Papapantoleon	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This is a compulsory course of the Financial Engineering Specialization. Students need to be familiar with probability theory and analysis. Knowledge of discrete time stochastic processes is beneficial, but not necessary. Students from other faculties are welcome, no need to write in advance, but they should be aware that this is a highly formalized course, belonging to the Master of Applied Mathematics	
Course Contents	The aim of this course is to introduce students to financial derivatives and the mathematics of finance. The course starts with basic financial derivatives (futures, options swaps) and then introduces the probabilistic modeling and study of financial markets. Topics covered include: no-arbitrage theory and the fundamental theorems of asset pricing, risk-neutral pricing of derivatives, discrete-time models, the binomial asset pricing model, American and exotic derivatives and the Black-Scholes model.	
Study Goals	At the end of the course, the students will know: - the basics of financial derivatives - the fundamentals of asset pricing in discrete time - modeling financial markets in discrete time - option pricing and hedging in discrete time - valuation of exotic and American derivatives - the Black-Scholes model	
Education Method	Lectures and possibly case studies.	
Literature and Study Materials	Föllmer & Schied: Stochastic Finance: An Introduction in Discrete Time, de Gruyter, 2016	
Prerequisites	This course is compulsory for all the Applied Maths students in the Financial Engineering Specialization. For all the others the course can be chosen as an elective, but it must be stressed that this is a rather mathematical course. The treatment of financial topics is at a high level of formalization.	
Assessment	Written exam (70%) + Assignments (30%). If the Covid-19 restrictions allow it, on-campus; otherwise, it will be an online examination, open book, with face-to-face remote check	
Remarks	This information is based on the "normal" situation; Corona-prevention measures may force changes affecting the way this course is taught.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Computational Science and Engineering AM 2022

Introduction 1

Specialisation Computational Science and Engineering

Compulsory orientation course:

WI4201 Scientific Computing

Compulsory specialisation courses (choose 3 out of 4 courses):

WI4204 Advanced Modeling in Science

WI4205 Applied Finite Elements

WI4475 Data Assimilation

WI4019 Nonlinear Differential Equations

Recommended specialisation courses:

WI4011-17 Computational Fluid Dynamics

WI4226 Advanced Systems Theory

WI4207 Continuous Optimization

WI4212 Advanced Numerical Methods

WI4450 Special Topics in Computational Science and Engineering

WI4635 Linear Algebra and Optimisation for Machine Learning

WI4209 Systems and Control

WI4221 Control of Discrete‐Time Stochastic Systems

IN4049TU Introduction to High Performance Computing

WI4771TU Object Oriented Scientific Programming with C++

WI4260TU Scientific Programming for Engineers

WI4655 Perturbation and Variational Methods for PDE

MM Inverse Problems in Imaging

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 3 out of 4)

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced , analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	Take home exam and oral exam. The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. The take-home exam and the oral exam count both for 50% in the final grading of the course.	
Special Information	disclaimer: information may change depending on the developments around the coronavirus. the study guide information may change depending on the developments around the coronavirus.	

WI4204	Advanced Modeling	6
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	2e semester intensieve cursus	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The participants will benefit from a solid knowledge of advanced calculus, basic knowledge of ordinary and partial differential equations and programming experience with Matlab (or an equivalent language).	
Course Contents	This modelling course is introducing the principles and practice of mathematical modelling. An important part of the course is to recognize the essential mechanisms governing a phenomenon. These mechanisms have to be translated into mathematical equations and formulas, and included into a mathematical model. This activity requires both a good understanding of the system under consideration and good mathematical skills. Mathematical modelling is essential for understanding processes in real life, complementing expensive experiments that sometimes cannot even be carried out at all. This course will concentrate on applications in physics and life sciences and focus on models in terms of partial differential equations. The course focuses on working in a small group on a specific project. The topics for project work may differ from year to year. For the academic year 2020-2021 some emphasis will be on continuum mechanics	
Study Goals	General 1. Has knowledge of mathematical modelling tools that are used for model applications in physics and for mathematical models formulated in terms of partial differential equations. 2. The student is able to recognize the principles and practice of mathematical modelling. 3. The student is able to identify the essential mechanisms governing a physical phenomenon and is able to translate these into mathematics and can include them into a mathematical model. Specific 1. Is able to make a partial differential equation non-dimensional and can apply scaling techniques to a partial differential equation. 2. Understands conservation principles and constitutive relations. 3. Is able to linearise partial differential equations in order to perform a stability analysis 4. Is able to develop a mathematical model for a practical problem and has the knowledge and maturity to analyse this model with the available mathematical tools. 5. Has the knowledge and skills for the use and assessment of mathematical models and is able to adapt the models for a specific application. 6. Is able to pose adequate questions and has a critical yet constructive attitude towards analysing and solving complex mathematical modelling problems 7. Is able to communicate verbally and in writing about mathematical research with specialists, non-specialists and other parties involved. 8. Has sufficient communicative skills to collaborate with others and discuss mathematical modelling problems and results.	
Education Method	The first lecture summarizes the mathematical modelling tools that might be useful for the projects. Furthermore during this lecture the projects are introduced by the supervisors and the groups of 2-4 students are formed. The main part of the course is offered as a one week "intensive course", fully devoted to the project. In most cases a computer program for simulating a phenomenon from physics of the life sciences has to be developed. At the end of the fourth quarter the written report of the project has to be handed in and the final presentations are scheduled.	
Literature and Study Materials	Many of the students will have a good background in continuum mechanics. The book Continuum Modeling in the Physical Sciences, E. van Groesen and J. Molenaar, Mathematical Modeling and Computation, Vol. 13, SIAM, 2007 (ISBN 978-0-898716-25-2) can be used as background material. Copies of this e-book can be borrowed from the TU-Delft library.	
Assessment	At the first lecture home assignments are given. The final deadline for submitting the home work is the start of the intensive week. The grading will be based on the individual homework (20%) and the project (80%). The grade of the project is based on the written report of the group work and on the oral presentation about the project. The following aspects of the project work will have equal weights: -The development of the model -Mathematical depth -Report and presentation disclaimer: information may change depending on the developments around the coronavirus.	
Pop-up tekst bij intekenen	Welcome to Advanced Modelling!	

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (50% of final grade) and a written exam (50% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. - No resit option will be available for the assignments. * Exam: - The written exam must be completed with a grade greater than 5.5. - Resit option will be available for the exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4475	Data Assimilation	6
Responsible Instructor	Prof.dr.ir. M. Verlaan	
Instructor	Prof.dr.ir. A.W. Heemink	
Contact Hours / Week x/x/x/x	0/0/2/0 lecture and 0/0/0/2 group work	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Partial differential equations and some knowledge on statistics	
Course Contents	Data assimilation methods are used to combine the results of a large scale numerical model with the measurement information available in order to obtain an optimal reconstruction of the dynamic behavior of the model state. An well-known example is weather forecasting where the numerical model of the atmosphere is continuous updated using the data that becomes available. The course gives an overview of the various methods used for data assimilation and discusses the underlying mathematical aspects.	
Study Goals	Students will understand the theory of data assimilation, will have an overview of the various techniques that are available, and will be able to use some of these techniques in practical applications. Specific Study goals 1. The student is able to rewrite a mathematical model in state space representation. 2. The student is able to formulation the data assimilation problem. 3. The student is familiar with variational data assimilation. 4. The student is able to solve parameter estimation problems using variational data assimilation. 5. The student is familiar with Kalman filtering for large scale systems 6. The student is able to implement and use ensemble modelling. 7. The student is able to implement and apply recursive data assimilation using ensemble Kalman filtering.	
Education Method	Lectures, homework, and a project. The course consists of seven 2-hour lectures. There are two homework sets. The project is handed out at the last lecture and continues until June.	
Literature and Study Materials	The following book will be used: Dynamic Data Assimilation: A least squares approach, John M. Lewis, S. Lakshmivarahan and S.K. Dhall, Encyclopedia of Mathematics and its Applications, Cambridge University Press, 2006.	
Assessment	The final grade consists of the following parts: * assignments (20% of final grade); * report about the project (50%); * oral exam (30%).	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations; some experience with a programming language (e.g., C) is preferred but not required).	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas is considered. The scaling behaviour of these algorithms in case of an increasing problem size and/or an increasing number of processors, is analysed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analysing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get a high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimisation for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found in the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Knowledge about high performance computer systems including parallel and distributed architectures, and programming models; 2. Basic knowledge about the concepts of data decomposition and parallel algorithms; 3. Knowledge about various high performance (numerical) methods and their parallelization; 4. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 5. Obtain some experience on performance analysis of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	Written exam (50%) + Lab work (50%).	
	disclaimer: information may change depending on the developments around the coronavirus.	
Enrolment / Application	Via Brightspace	

WI4011-17	Computational Fluid Dynamics	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor calculus * linear algebra * basic concepts of the finite element method * basic concepts of iterative solvers and preconditioning The student is also expected to be familiar with programming (e.g., in MATLAB or Python).	
Course Contents	Part I: Numerical methods for the convection-diffusion equation; Stability, consistency and convergence of the numerical methods; Extensions to incompressible Navier-Stokes equations; Part II: Basic iterative solvers and preconditioners for the Poisson problem Schwarz domain decomposition preconditioners for the Poisson problem Extension of Schwarz preconditioners to saddle point problems (Stokes and Navier-Stokes equations)	
Study Goals	Learning objectives of the course: Upon completion, the student should be able to: Part I: 1. Derive the equations of fluid dynamics using multivariable calculus and physical conservation laws 2. Explain the singularly-perturbed behavior of solutions to the equations of fluid dynamics 3. Discretize the convection-diffusion equation and derive discretization errors 4. Choose discretizations suitable for convection-dominated flows 5. Discretize the incompressible Navier-Stokes equations Part II: 6. Construct and implement Schwarz domain decomposition methods 7. Explain the concept of the abstract theory of Schwarz methods 8. Apply the abstract Schwarz theory to one- and two-level Schwarz preconditioners to derive condition number bounds for the Poisson problem 9. Apply nonlinear and linear solution techniques to (Navier-)Stokes equations 10. Extend Schwarz methods to (Navier-)Stokes equations	
Education Method	The course comprises 14 lectures (7 lectures for Part I taught by S. Jain, 7 lectures for Part II taught by A. Heinlein).	
Course Relations	In the past WI4011 briefly covered a number of subjects more extensively discussed in the course WI4201 'Scientific Computing'. This overlap has been removed, and therefore student are strongly recommended to take the course WI4201.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * "The Finite Element Method" by TJR Hughes * "Domain Decomposition Methods - Algorithms and Theory" by A. Toselli and O. B. Widlund * "Domain Decomposition Methods for Partial Differential Equations" by A. Valli and A. Quarteroni Additional current topics, which are not covered in standard text books yet, will also be discussed. Further references will be given during the course.	
Assessment	Course will be assessed based on individual assignments (50% of final grade) and project work (50% of final grade; project report and presentation are graded). To pass the course, the following is mandatory. * The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. * The projects must be completed with grades greater than 5.5. No resit option will be available for the assignments, only for the final project.	
Remarks	Disclaimer: Information may change depending on the developments around the coronavirus. This course focuses on the mathematical background of numerical methods for the incompressible Navier-Stokes equations and convection-diffusion equation. For the non-mathematician, it will show how many aspects of the methods can be mathematically analyzed and motivated, far beyond what is shown in more engineering focused courses. For the mathematician, it will show how his knowledge can be applied in one of the most challenging fields of mathematical/computational modeling.	
Co-Instructor	Dr.ir. S. Jain	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid bachelor level knowledge linear algebra and multivariate analysis. The student should also have knowledge of linear optimization and convex analysis to the level of being able to follow the text and do the exercises from the following: Linear Programming, A Concise Introduction, Thomas S. Ferguson: Available at https://www.math.ucla.edu/~tom/LP.pdf Chapters 1 and 2, along with the accompanying exercises. Convex Optimization, Stephen Boyd and Lieven Vandenbergh: Available at http://stanford.edu/~boyd/cvxbook/ Sections: 2.1, 2.2 and 3.1. Exercises (from the book): 2.1, 2.2, 2.12, 3.1, 3.3, 3.5 and 3.7	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be constrained by (differentiable) equality and inequality constraints as well as by convex cone constraints. Optimization problems like this occur naturally and commonly in science and engineering and also occur as relaxations of discrete optimization problems. Differentiability of the functions defining the problems allows for the use of multivariable calculus and linear algebra techniques to study the problems and to design and analyze efficient algorithms. This course aims to provide a concise introduction into the basics of continuous unconstrained, constrained and conic optimization.	
Study Goals	The student will be able to: - Prove results on (convex) optimization problems. - Solve the KKT conditions for basic constrained optimization problems. - Be able to formulate the Lagrange dual, and understand/prove basic results on these problems. - Give both sufficient and necessary optimality conditions for constrained continuous optimization problems. - Use a range of techniques to solve both unconstrained and constrained continuous optimization problems, and prove results on these techniques. - Formulate and recognize conic optimization problems, along with being able to construct their dual problems.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Remarks	This is a LNMB Mastermath course taught by Daniel Dadush.	

WI4209	Systems and Control	6
Responsible Instructor	Dr. J.W. Polderman	
Contact Hours / Week x/x/x/x	1st semester intensive course Contact for students: j.w.polderman@utwente.nl	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The course aims at students in pure and applied mathematics with an interest in applications of (linear) algebra and ordinary differential equations. The purpose of the course is to introduce the students to basic concepts and more advanced notions of the mathematical theory of systems.	
Course Contents	The main objects of study in this course are systems modeled by linear time-invariant differential equations. We start with a treatment of the theory of algebraic representation of dynamical systems using polynomial matrices. The main tool is the Euclidean algorithm applied to matrices of real polynomials. The main result is a complete characterization of all representations of a given system. Several other representation are introduced along with their relations. Important examples of such representation are input-output representations that reveal that some variables may be unrestricted by the equations, and state space representations that visualize the separation of past and future, also referred to as the Markov property. Controllability and observability are important system theoretic concepts. A controllable system has the property that a desired future behavior can always be obtained, independent of the past behavior, provided that this future behavior is compatible with the laws of the system. Observability means that the complete behavior may be reconstructed from incomplete observations. The theory of controllability and observability forms one of the highlights of the course. Stability can be an important and desirable property of a system. Stabilization by static or dynamic feedback is one of the key features of Systems and Control. In the pole placement theorem linear algebraic methods and the notion of controllability are used in their full strength. The theorem, loosely speaking, says that in a controllable system the dynamic behavior can be changed as desired, in terms of characteristic values, by using appropriate feedback. It forms one of the most elegant results of the course and indeed of the field of Systems and Control.	
Study Goals	Mathematical systems theory is concerned with problems related to dynamic phenomena in interaction with their environment. These problems include: Modeling. Obtaining a mathematical model that reflects the main features. A mathematical model may be represented by difference or differential equations, but also by inequalities, algebraic equations, and logical constraints. Analysis and simulation of the mathematical model. Prediction and estimation. Control. By choosing inputs or, more general, by imposing additional constraints on some of the variables, the system may be influenced so as to obtain certain desired behavior. Feedback is an important example of control.	
Education Method	Lectures, exercises, self-study and intensive week	
Literature and Study Materials	Introduction to Mathematical Systems theory: a Behavioral Approach, by J.W. Polderman and J.C. Willems (Springer, New York, 1998). Additional lecture notes will be handed out during the course. NOTE: The paper version of the book mentioned above is sold out. Fortunately, the book is available as eBook. For more details, check the website of Springer: http://www.springer.com/mathematics/book/978-1-4757-2955-9	
Prerequisites	The course is aimed at students in mathematics at the comprehensive as well as the technical universities. Solid knowledge of linear algebra and calculus is essential. Furthermore, knowledge of ordinary differential equations is desirable. All at the bachelor level.	
Assessment	The final grade is determined by the results of the homework assignments, the oral presentation, the written report and a written exam. Details will be made available in due time.	
Remarks	disclaimer: information may change depending on the developments around the coronavirus. This is an intensive course in the 2019/2020 Mastermath programme. For information about the organization see at http://elo.mastermath.nl/	
Co-Instructor	Dr. J.W. van der Woude	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	In the first period a set of exercises is given. These should be worked out, strict deadline 26-3-2019 (Grade: G_1). The exercise should be handed in individually. In the middle of April a take home exam including practical exercises is given. This exam can be done by groups of two students. The report of this exam should be returned to us before 11-6-2019 (Grade: G_2). Thereafter an appointment can be made for an oral examination over Chapter 18 (including exercise 18.1, 18.2, and 18.3), 19, and 20 (Grade: G_3). Note that material discussed in Chapter 1 to 9, which is relevant for the material in Chapter 18 to 20 can also be part of the oral examination. The oral exam must be passed with a 6 or higher. Then, provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 G_3)/4$. disclaimer: information may change depending on the developments around the coronavirus.	

WI4221	Control of Discrete-Time Stochastic Systems	6
Responsible Instructor	Prof.dr.ir. J.H. van Schuppen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Control and system theory (undergraduate level), linear algebra, differential equations, stochastic processes.	
Course Contents	Discrete-time stochastic systems, distributions and invariant measures. Stochastic realization. Control with complete observations, optimal control theory, dynamic programming for finite and infinite horizons. Kalman filtering and special cases of filtering of stochastic systems. Control with partial observations, separation property. Control of networked stochastic systems and decentralized control.	
Study Goals	Students will be able to explain the fundamental concepts of stochastic systems. They will be able to solve elementary optimal control problems of stochastic systems by dynamic programming. They will understand the derivation of the Kalman filter. Finally, they will be able to explain the control of stochastic systems with partial observations.	
Education Method	Lectures/exercises	
Literature and Study Materials	Notes for this course can be obtained from the instructor.	
Assessment	Oral exam based on lecture notes and on home work sets. Course grade based on oral exam (50%) and homework sets (50%). Home work sets cannot be redone. The oral exam can be done only once. disclaimer: information may change depending on the developments around the coronavirus.	
Elective	Yes	
Tags	Mathematics Signals and Systems Stochastics	

WI4226	Advanced System Theory	6
Responsible Instructor	Dr. J.W. van der Woude	
Contact Hours / Week x/x/x/x	0/0/4/0 hc	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	The BSc course Systems Theory AM2530, or an equivalent course, and a good knowledge of basic mathematics.	
Course Contents	Part I In part I of this course the connection of linear system theory and convex optimization is illustrated. One of the key ingredients are the so-called Linear Matrix Inequalities, LMIs for short. LMIs can be treated efficiently by means of semi-definite programming techniques coming from convex optimization. It turns out that many properties of linear systems, like stability, controllability, observability, etc., can be formulated in terms of LMIs. Also the design of controllers satisfying stability and other constraints can be done efficiently using semi-definite programming and LMIs. The first part of part I starts by recalling basic knowledge from linear system theory and placing it in the frame work of LMIs. A second topic in part I will be LQ optimal control and the introduction of dissipativity. Both topics are of crucial importance for system theory. LQ optimal control has a long and rich history, but is still important and applicable. Dissipativity also has a long history, but its applicability has increased in recent years by the event of new efficient algorithms to solve semi-definite programming problems. The last topic in part I are system norms and the design of a controller such that the combined system behaves in a desired way specified in terms of its norm. To that end, the H-infinity and the H2 norm will be introduced. Also methods will be treated by which it can be investigated whether a certain desired norm can be achieved, and how this then actually can be done by means of state or output feedback. Part II A short introduction to general optimal control theory, containing stability issues for nonlinear systems, calculus of variations, the maximum principle, and so on. The remaining time of part II will be reserved to study a document that sheds some additional light on traditional approaches and methods that are mostly treated in regular system theory courses. The students are requested to study a relevant document and write a report on it. The document and topic can be different each year.	
Study Goals	Advance system theory deals with the extensions in/of linear system theory. After the course the student will be familiar with the following topics. - Convex optimization and linear matrix inequalities - LQ optimal control and dissipativity - Controller design and the H_{∞} norm/H_2 norm - Introduction to general optimal control theory - An additional document concerning some well-known results from systems theory	
Education Method	Lectures, take home exercises, self study in a small project on some relevant document.	
Literature and Study Materials	Lecture notes will be made available at the beginning of the teaching period through Brightspace. Homework sets, slides used during the lectures and additional material become available during the teaching period.	
Assessment	Grade for course will be weighted average of both written exams ($2 \times 20\%$), the average of all homework exercises ($1 \times 40\%$) and the grade for the report ($1 \times 20\%$). A resit over the two written term exams will be scheduled on an individual basis.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Although a starting point, the above Course Contents may be adapted slightly during the course depending on the knowledge of the students. The second part of the Course Contents and the last topics mentioned as Study Goals are tentative and may be subject to change. Also the setup of the second part may be subject to changes.	

WI4260TU	Scientific Programming for Engineers	3
Responsible Instructor	Dr.ir. H.X. Lin	
Practical Coordinator	Ir. C.W.J. Lemmens	
Contact Hours / Week x/x/x/x	0/0/2/0 + lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Experience with a programming language (e.g., Python or C), linear algebra, experience with linux is a plus (but not a prerequisite).	
Course Contents	The course tries to bring students to a level where they are able to change algorithms from e.g., numerical analysis into efficient and robust programs. It comprises: 1. Introduction to programming in general; 2. Floating point number rounding-off errors and numerical stability; 3. (Numerical) Software design; 4. Data Structures; 5. Testing, debugging and profiling; 6. Efficiency issues in computing time and memory usage; 7. Optimization and dynamic memory allocation; 8. Scientific software sources and libraries. P.S. (i) This is not a general introductory course to learn how to write a computer program. Furthermore, the course concentrates mainly on sequential programming and only briefly introduces parallel programming (MPI and OpenMP). More advanced topics like threads or parallel (MPI/GPU) programming on supercomputers are not covered by this course (they are covered by other courses, e.g., Introduction to High Performance Computing). (ii) This is a course aimed for master students, however, if you have had followed the course Scientific Programming in the BSc CSE-minor, then you should not select this course as a Master elective course due to the large overlap.	
Study Goals	1. Learn how to program in a high level programming language; 2. Learn about the problems related to the use of floating-point numbers in scientific computing; 3. Can make the transition from scientific model to a structured program; 4. Obtain basic knowledge about Optimization, Debugging and Profiling of these programs.	
Education Method	Weekly there are 2-hour lectures and 3-hour lab sessions. Participation in a minimum of five lab sessions is required, with the optional homework, a bonus of max. 1 point can be obtained.	
Books	Textbook: Writing Scientific Software - A guide to good style, by Suely Oliveira and David Stewart, Cambridge University Press, ISBN-13 978-0-521-67595-6	
Assessment	There is a 3-hours exam comprising theory and practical (lab) questions. The final grade is determined by 90% exam score + 10% lab homework.	
Tags	disclaimer: information may change depending on the developments around the coronavirus. Numeric Methods Programming	

WI4450	Special Topics in Computational Science and Engineering	6
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course discusses the state-of-the-art of a selected topic from computational science and engineering. This topic changes yearly. The first part of the course gives an introduction into the subject through lectures in Q3. In the second part of the course (in Q4) a project will be carried out to apply the theory and methods taught in the first part of the course. The topic of the course of year 2022/2023 will be on Preconditioned Krylov Methods for High Performance Computing. It will be given by Jonas Thies and Martin van Gijzen Modern supercomputers like the DelftBlue consists of thousands of cores. In order to make use of the enormous computing power of these computers specialised algorithms are required. This course focusses on modern Preconditioned Krylov Methods for High Performance Computing. These methods are used for solving large scale linear algebra problems like solving of linear systems and eigenvalue problems. We review a number of Krylov Methods and discuss how they can be adapted for parallel computing. We will also review state-of-the-art preconditions for parallel computing, for example based on domain decomposition. In Q4 some external experts will give lectures about their recent research.	
Study Goals	At the end of the course the student: Will have a good overview of the current research on the course topic; Is able to read and understand the recent literature; Is able to perform research on the topic.	
Education Method	Introductory lectures, lectures by external specialists, project	
Literature and Study Materials	Lecture notes, relevant research papers and a list of recommended background material will be made available via Brightspace.	
Prerequisites	The course WI4201 is prerequisite knowledge. Programming in C or Fortran is required. We will give you a short tutorial to help you get on your way, but the material taught in the courses WI4260TU Scientific Programming for Engineers and WI4771TU Object Oriented Scientific Programming with C++ will be helpful and these courses are therefore recommended.	
Assessment	Project presentation (50%) and report (50%)	
	disclaimer: information may change depending on the developments around the coronavirus.	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will deal with machine learning from the perspective of linear algebra and optimization. In particular, we will discuss how different machine learning techniques use classical concepts from linear algebra and optimization and how their performance can be improved by using more sophisticated linear algebra and optimization algorithms. We cover theoretical aspects as well as the implementation and application of these techniques. Among others, we will study + efficient matrix operations such as matrix decompositions and factorizations, + graph-related algebra, + first- and second-order optimization algorithms with their learning-oriented customization and regularization techniques, and + optimization on computational graphs (backpropagation).	
Study Goals	The student has gained an understanding of how linear algebra and optimization algorithms are used in and how these are important for the performance and efficiency of different machine learning techniques: clustering, classification and regression, neural networks, matrix decomposition/factorizations Furthermore, the student is able to + motivate and implement different matrix decomposition methods + analyze and explain matrix factorization methods + analyze graph-based data using linear algebraic tools + implement and customize first- and second- order unconstrained optimization methods + analyze and implement constrained and duality-based optimization algorithms + analyze and efficiently implement forward and backward propagation through neural networks	
Education Method	Lectures	
Assessment	Group assignments/projects and a written exam	
Co-Instructor	Dr. K.S. Postek	

WI4655	Perturbation and Variational Methods for Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	The course consists of two part. In the first part of the course (Q1), methods will be introduced to get approximate solution to PDEs. In the lectures we elaborate on asymptotic approximations, matched asymptotic expansions, multiple scales, WKB method, bifurcation and stability analysis. In the second part (Q2), a modern introduction to the field of elliptic PDEs is given. In the lectures we consider weak solutions to elliptic equations (such as Laplace equation), Sobolev spaces. Modern PDEs uses tools from functional analysis and measure theory. The precise regularity and behavior of solutions of elliptic equations is one of the main features in the lectures. For non-linear PDEs such results are fundamental. The course might be useful for student in analysis, PDE and numerics, or anyone who wants to know more about Sobolev spaces and second order equations. (Key words: elliptic, weak derivatives, Sobolev spaces extension theorem, trace theorem, Sobolev inequalities, Hilbert techniques, compactness theorems, second order elliptic equations, Weak solution, energy estimates, regularity, eigenvalue techniques).	
Study Goals	Global objectives for the first part of the course: After following this course the student will be able to apply fundamental and advanced analytical techniques and methods to study properties of solutions of ordinary and partial differential equations. The mathematical methods and techniques that the student is able to use are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations. Global objectives for the second part of the course: After the course the student - knows what a weak derivative is - understands the definition of L^p and Sobolev spaces. - can apply basic Sobolev space techniques (approximation, extension, trace, Sobolev embedding, compactness) - can give the definition of a weak solution of a second order elliptic PDE - can apply Sobolev space techniques to prove existence and uniqueness of weak solutions of second order elliptic PDEs - can derive properties (regularity/estimates) of solutions of second order elliptic PDEs Objective per topic: The student Sobolev spaces the student -knows the definition of Hölder spaces -knows the definition of the weak derivative -can calculate weak derivatives -knows the definition of Sobolev spaces -can approximate Sobolev functions by smooth functions -can explain the extension theorem -can explain the trace theorem -knows Sobolev inequalities -can apply Sobolev inequalities -can derive Sobolev inequalities -knows the compactness theorem -can apply the compactness theorem -understands what the dual space of a Sobolev space is Second order elliptic equations the student -knows the definition of a weak solution. -can give energy estimates -can apply the Lax-Milgram theorem to obtain existence of weak solutions -knows what regularity is -can state regularity results for second order equations (interior and boundary) -can apply eigenvalue techniques.	
Education Method	First part of the course: Introduction to Perturbation Methods (second edition) by M.H. Holmes Second part of the course: lecture notes. Optional literature: Partial Differential Equations, by Lawrence C. Evans, AMS, 1998, 662 pp., Hardcover, ISBN 0-8218-0772-2. Assessment	
Assessment	Two partial exams, one at the end of each period. Minimum of a 5 for each partial exam, the final grade is the average of both grades.	
Co-Instructor	Dr.ir. M.C. Veraar	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). Final project (group work) starting mid Q2 with submission deadline end Q2 / beginning Q3 (2/3 of final grade).	
Co-Instructor	K. Cools	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Financial Engineering AM 2022

Introduction 1

Specialisation Financial Engineering

Compulsory orientation course:

WI4675 Introduction to Financial Mathematics

Compulsory specialisation courses:

WI4430 Martingales, Brownian Motion

WI4201 Scientific Computing

WI4079 Financial Mathematics

WI4154 Computational Finance

Recommended specialisation courses:

WI4425 Financial Markets Theory

WI4505 Quantitative Risk Management

WI4635 Linear Algebra and Optimisation for Machine Learning

WI4230 Time series and Extreme Value Theory

WI4052 Risk Analysis

MM Nonparametric Statistics

WI4614 Stochastic Simulation

WI4019 Nonlinear Differential Equations

WI4156(TU) Game theory

WI4205 Applied Finite Elements

WI4630 Statistical Learning

WI4771TU Object Oriented Scientific Programming with C++

WI4224 Special Topics in Financial Engineering

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4079	Financial Mathematics	6
Responsible Instructor	Dr. A.F.F. Derumigny	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This is a compulsory course of the Financial Engineering Specialization. For all the others the course can be chosen as an elective, but it must be stressed that this is a rather mathematical course. Therefore, students from other faculties are welcome, no need to write in advance, but they should be aware that this is a highly formalized course, belonging to the Master of Applied Mathematics: the treatment of financial topics is at a high level of formalization.	
Course Contents	Students need to be familiar with the contents of "Martingales, Brownian Motion, and Stochastic Processes" (WI4430). It is beneficial, but not mandatory, that students have followed the course "Introduction to Financial Mathematics" (WI4675). The aim of the course is to introduce the students to the probabilistic study of the classical theory of finance. The first part of the course is centered around stochastic calculus. We will then cover arbitrage theory, perfect markets, risk-neutral pricing of options and other derivatives, Black-Scholes-Merton model, models for exchange rates and interest rates. For each topic and model we will stress its points of strength, but also its points of weakness.	
Study Goals	At the end of the course, the students will know: - the basics of Ito calculus; - the fundamentals of asset pricing; - how to change measure and the impact of such an operation in pricing and financial mathematics; - the importance of risk-neutrality; - how to price the most important derivatives; - how to deal with some of the most important issues in risk management (VaR, error estimation, loss estimation); - how basic exchange and interest rates work and how they can be efficiently modeled.	
Education Method	Lectures and possibly case studies.	
Literature and Study Materials	TBD. However, the book by Shreve (2004), Stochastic calculus for finance II: continuous models (Springer, New York) gives a good indication of contents and level of the course.	
Assessment	Written exam (100%). If the Covid-19 restrictions allow it, on-campus (without any books or notes); otherwise, it will be an online examination, open book, with face-to-face remote check.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4154	Computational Finance	6
Responsible Instructor	F. Fang	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of Ito's lemma, partial differential equations (PDEs), linear algebra and the basic of numeric analysis such as numerical integration rule, computing tool: Python.	
Course Contents	The course aims at introducing the basic option pricing theory and various computational methods to price options in practice. More precisely, the derivation of pricing PDE and Feynman-Kac theory, derivation of the characteristic function for affine processes, Finite difference method to solve the PDE numerically, Monte-Carlo method and Fourier method to price the options. An overview of the financial products traded nowadays will also be covered.	
Study Goals	For different mathematical models, for products from the financial industry, different numerical mathematics techniques are required. The students learn about the numerical techniques used in finance. Finite differences for PDEs, Monte Carlo techniques for complicated integrals and Fourier-based integration are taught for the numerical side; For the financial product modelling side, we discuss asset pricing processes and a variety of options, from basic European options to options with early-exercise opportunities and so-called exotic options. Furthermore, details about financial products are given.	
Education Method	Lectures	
Literature and Study Materials	The course will be taught based on slides and some background material. The book "Mathematical Modeling and Computation in Finance", by C.W. Oosterlee and L.A. Grzelak, World Scientific Press, 2019, is used as background information.	
Assessment	There are two sets of exercises, which together count for 20% of the final score. One intermediate take-home exam will take place in the middle of the course, and will count for 30% of the final score. There will be a final, oral exam at the end, which accounts for 50% of the final score.	
Remarks	The course is linked to WI4079 (Financial Mathematics), where the stochastic concepts of finance are introduced.	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5.	
Co-Instructor	disclaimer: information may change depending on the developments around the coronavirus. Prof.dr.ir. M.B. van Gijzen	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	Two graded homeworks and a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f \geq 5$. disclaimer: information may change depending on the developments around the coronavirus. The homework grade does not count for the resit.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced , analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	Take home exam and oral exam. The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. The take-home exam and the oral exam count both for 50% in the final grading of the course.	
Special Information	disclaimer: information may change depending on the developments around the coronavirus. the study guide information may change depending on the developments around the coronavirus.	

WI4052	Risk Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics	
Summary	Risk analysis course will treat problems often appearing in applications. Some generic methods that are applied in many different areas of applications will be discussed. We will not concentrate on any particular application, rather we will discuss important assumptions made in different models and their applicability for different real life problems. Moreover we show how parameters of these models can be obtained from data and discuss methods of validation of the obtained model.	
Course Contents	Risk modelling, life distributions, event trees, fault trees, Bayesian Belief Nets, reliability data bases, dependence modelling, software reliability, expert judgment.	
Course Contents Continuation	Introduction & Uncertainty Probability Multivariate, Copula Stochastic processes Bayesian, classical estimations & Testing Fault trees Bayesian belief nets Dependent failures Expert Judgment Reliability data bases Software reliability	
Study Goals	Students must be able to justify use of probabilistic methods in risk analysis and apply techniques like fault trees and Bayesian belief nets. They should know how to estimate parameters of models for dependent failure rates. Moreover they should be able to perform competing risks analysis for simple reliability data sets. Students should understand basic techniques of incorporating expert knowledge into risk models.	
Study Goals continuation	The goal of the course is to expose students to problems appearing in applications. During lectures the theory will be treated and exercises will be provided. Students will be required to complete the assignment where some practical problems have to be solved. The theoretical questions are not required. However making them allows to get a feedback and extra explanation in how they should be solved. These exercises together with course material will constitute basis for the exam in January. Practical exercises are required and will form part of your final grade.	
Education Method	Lectures and Assignments	
Literature and Study Materials	Parts of the book "Probabilistic Risk Analysis", Bedford & Cooke, Cam.U. press, 2001. Extra material on the Brightspace	
Reader	Brightspace	
Prerequisites	Basic probability and statistics	
Assessment	Written.	
Permitted Materials during Tests	disclaimer: information may change depending on the developments around the coronavirus. Calculator and self made A4 with formulas.	
Tags	Mathematics Stochastics	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr.ir. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Assessment	written exam	
	disclaimer: information may change depending on the developments around the coronavirus.	

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (50% of final grade) and a written exam (50% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. - No resit option will be available for the assignments. * Exam: - The written exam must be completed with a grade greater than 5.5. - Resit option will be available for the exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4224	Special topics in Financial Engineering	6
Responsible Instructor	Dr. A. Papapantoleon	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course will deal with current topics in financial mathematics and financial engineering, such as the applications of machine learning methods in finance.	
Study Goals	- To understand the applications of machine learning methods in mathematical finance - To know the strengths and the weaknesses of all the discussed models. - To acquire useful and practical knowledge in these methods.	
Education Method	Lectures from the instructor and presentations from the students.	
Literature and Study Materials	M. F. Dixon, I. Halperin, P. Bilokon: Machine Learning in Finance: From Theory to Practice. Springer, 2020.	
Prerequisites	"Financial Mathematics" (WI4079) and "Computational Finance" (WI4154).	
Assessment	The presentations from the students will be graded.	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	The final assessment will be oral or written, depending on the number of the students registered. (If there are at least 10 students we will organize a written exam.) There will be a final assessment for the extreme value theory part at the end of Q3 and a final assessment for the time series part at the end of Q4. In summer there will be the opportunity for a resit in the form of an oral exam for each part separately. There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory	

0.35 exam extreme value theory
0.15 average over assignments time series
0.35 exam time series

WI4425	Financial Markets Theory	6
Responsible Instructor	Dr. J.H.M. Anderluh	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Choices under Uncertainty, General Equilibrium Theory, Pareto Optimality, Expected Utility Theory, Risk Aversion Choices under risk : Portfolio Problem, Insurance Demand en Prudence, Stochastic Dominance, Mean Variance Analysis Portfolio frontier, Mutual Funds separation, Risk sharing and Pareto optimality Asset Markets, Intertemporal Consumption, the F. theory of asset pricing, CAPM CAPM, APT, Portfolio Choice Multi Period Markets Information and Financial Markets Uncertainty, Rationality and Heterogeneity Financial Markets Microstructure Corporate Finance	
Study Goals	The main goal is to enable the student to quantify and in some cases optimise the relation between return and risk in a financial markets situation. The student will be able to build a mathematical model of the situation at hand with ingredients like equilibrium concepts, no-arbitrage concepts.	
Education Method	One block of 2 hours lectures every week during the 1st semester. A written exam afterwards.	
Books	Emilio Barucci, Financial Markets Theory	
Assessment	Written exam	
disclaimer: information may change depending on the developments around the coronavirus.		

WI4505	Quantitative Risk Management	6
Responsible Instructor	F. Barsotti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The implementation of sound and reliable quantitative risk models is of vital interest for all banks and financial institutions, both from an economic and a legal point of view. This course provides a rather comprehensive treatment of the theoretical concepts and modeling tools of quantitative risk management. At the end of the course, students will master both theory and practice, and they will be able to solve real world problems. We will cover the modeling of important risks like: market risk, operational risk and model risk. All models will be analyzed in their points of strength and, most importantly, their points of weakness, to address the main deficiencies of current practice.	
Study Goals	- To understand risk, epistemologically, statistically and in practice. - To be able to use the main risk measures available and, in case, to develop new ones. - To know the basics of EVT. - To be able to model market, credit and operational risks, using the most recent techniques. - To be able to estimate most of the models we will discuss together on real data. - To know the strengths and the weaknesses of all the discussed models, i.e. to understand the importance of model risk. - To acquire useful and practical knowledge in risk management.	
Education Method	Front lectures, as necessary supplemented by online materials including videos, case studies.	
Literature and Study Materials	McNeil, Frey and Embrechts: Quantitative Risk Management, Princeton University Press, 2015.	
Prerequisites	"Financial Mathematics" (WI4079) and "Computational Finance" (WI4154).	
Assessment	Written / Oral exam (100%). If the Covid-19 restrictions allow it, on-campus; otherwise, it will be an online examination, open book, with face-to-face remote check.	
disclaimer: information may change depending on the developments around the coronavirus.		
Permitted Materials during Tests	Standard non graphical calculator.	
Special Information	If you are not a mathematics student, feel free to join the course, no need to ask, but be aware that this course is highly formalized and relies on good mathematical foundation knowledge. This is a course of the master in applied mathematics!	
Remarks	A proviso: it is the firm intention of the faculty to offer this course, but (as of June 2020) it is not certain whether this will actually be possible. A second one: This information is based on the "normal" situation; Corona-prevention measures may force changes affecting the way this course is taught.	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, and the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	Mini-projects (25%); final exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Essentially, a BSc program in mathematics is assumed. Specifically, knowledge, understanding and ability in calculus (AM1040, AM1070), probability (AM1080), statistics (AM2080), optimization (Lagrange multipliers) and linear algebra (AM1030, AM2010) at the level of the TU Delft BSc program in Applied Mathematics. The MSc courses Linear Algebra and Optimization for Machine Learning (WI4635) and Statistical Inference (WI4455) are highly recommended as prerequisites. Furthermore basic programming skills in Python are required (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Some measure theory (AM2090) may be helpful but will not be assumed.	
Course Contents	The overarching goal in this course is to develop methods for estimating (or 'learning') an unknown function from data or make predictions of unseen function outputs. In the course we will introduce key ideas, models, algorithms and theories related to the subject of statistical learning. The detailed contents of the course may vary but will include: - linear regression and logistic regression - avoiding overfitting in high-dimensional models (shrinkage methods, LASSO) - kernel methods and Gaussian process regression - support vector machines - model selection - neural networks and deep learning - stochastic gradient descent	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Upon successful completion of this course, the student 1. knows and understands the objective of statistical learning and the curse of dimensionality in the context of statistical learning; 2. can apply and analyse relevant models for statistical learning, including linear regression, logistic regression, sparse regression models, support vector machines and neural networks; 3. can apply and analyse ways for avoiding overfitting in the context of statistical learning; 4. can apply and analyse methods for extending, combining and selecting models, such as kernel methods and cross validation; 5. can apply and analyse computational methods and algorithms used in statistical learning, such as stochastic gradient descent.	
Education Method	Lectures, exercises, homework assignments, practice exam. Homework assignments will consist of theoretical exercises and computer assignments.	
Literature and Study Materials	The main reference will be: Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ Additional material is to be determined and will be made available during the course. Background reading / additional material: - Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ - Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). MIT press Cambridge. https://www.deeplearningbook.org/	
Assessment	assignments during the course (40%), final written exam (60%), sufficient grade (5.8) for both parts required	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will deal with machine learning from the perspective of linear algebra and optimization. In particular, we will discuss how different machine learning techniques use classical concepts from linear algebra and optimization and how their performance can be improved by using more sophisticated linear algebra and optimization algorithms. We cover theoretical aspects as well as the implementation and application of these techniques. Among others, we will study + efficient matrix operations such as matrix decompositions and factorizations, + graph-related algebra, + first- and second-order optimization algorithms with their learning-oriented customization and regularization techniques, and + optimization on computational graphs (backpropagation).	
Study Goals	The student has gained an understanding of how linear algebra and optimization algorithms are used in and how these are important for the performance and efficiency of different machine learning techniques: clustering, classification and regression, neural networks, matrix decomposition/factorizations Furthermore, the student is able to + motivate and implement different matrix decomposition methods + analyze and explain matrix factorization methods + analyze graph-based data using linear algebraic tools + implement and customize first- and second- order unconstrained optimization methods + analyze and implement constrained and duality-based optimization algorithms + analyze and efficiently implement forward and backward propagation through neural networks	
Education Method	Lectures	
Assessment	Group assignments/projects and a written exam	
Co-Instructor	Dr. K.S. Postek	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). Final project (group work) starting mid Q2 with submission deadline end Q2 / beginning Q3 (2/3 of final grade).	
Co-Instructor	K. Cools	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Discrete Mathematics and Optimization AM 2022

Introduction 1

Specialisation Discrete Mathematics and Optimization

Compulsory orientation course:
WI4227-14 Discrete Optimisation

Compulsory specialisation course:
WI4620 Semidefinite Optimization

Compulsory specialisation courses (choose 2 out of 4 courses):
WI4515 Relaxations and Heuristics
WI4670 Extremal Combinatorics
WI4410 Advanced Discrete Optimization
WI4635 Linear Algebra and Optimisation for Machine Learning

Recommended specialisation courses:
WI4207 Continuous Optimization
MM Scheduling
WI4630 Statistical Learning
MM Quantum Information Theory
MM Quantum Computing
MM Machine Learning Theory
MM Queuing Theory
ET4030 Error Correcting Codes
MM Algebraic Methods in Combinatorics

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4620	Semidefinite Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Semidefinite programming is a relatively recent tool in mathematical optimization and can be seen as a vast generalization of linear programming. One can define semidefinite programming as the minimization of a linear functional over positive semidefinite matrices subject to linear constraints. Only twenty years ago it became clear that one can solve semidefinite optimization problems efficiently. Since then semidefinite optimization has become a frequently used tool in mathematics of big expressive and computational power. The main focus of this course is the application of semidefinite programming techniques to mathematical problems, such as the independent set and graph coloring problems, the max-cut problem, Nesterov's approximation algorithm, Grothendieck's inequality, quantum information theory (no prior quantum knowledge assumed), and polynomial optimization. We will also discuss the duality theory of semidefinite programming including a proof of Slater's condition and the required convex analysis.	
Study Goals	See the Brightspace page of this course.	
Education Method	The course consists of lectures and a written exam.	
Assessment	The grade is based on a written exam. The resit is also a written exam.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

WI4410	Advanced Discrete Optimization	6
Responsible Instructor	Prof.dr.ir. K.I. Aardal	
Instructor	D. de Laat	
Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Linear optimization: basic solutions and their relation to extreme points, simplex method, duality and complementary slackness Integer optimization: basic cutting planes, branch-and-bound Computational complexity: Complexity classes P and NP and NP-complete, polynomial reductions	
Course Contents	Integer optimization: Polyhedral theory; Cutting planes; Gomory mixed-integer cuts, split cuts and their relation; Lift-and-project; IP in fixed dimension: Lattices and basis reduction; IP in fixed dimension: Lenstra's algorithm and extensions Parameterized complexity: fixed-parameter tractability, bounded search, kernelization, dynamic programming, treewidth Applications in phylogenetics: phylogenetic trees, agreement forests, SPR distance, phylogenetic networks	
Study Goals	To understand and be able to use the methods and concepts described in the course. The student should also be able to prove results similar to the ones described in the course.	
Education Method	There will be 2 hours of lectures per week. Exercises will be given that the students can do at home.	
Literature and Study Materials	The material will be made available to the students. It will consist of various scientific articles and textbook chapters.	
Assessment	Written exam	
disclaimer: information may change depending on the developments around the coronavirus.		

WI4515	Relaxations and Heuristics	6
Responsible Instructor	Dr.ir. J.T. van Essen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* Programming skills (Python/C/C++) * Linear algebra * Linear programming, simplex method, LP duality (covered in TW2020) * Integer linear programming, branch-and-bound (covered in TW2020) * Complexity theory	
Course Contents	Many real-life optimization problems are NP-hard. For these problems, it is not known whether there exist efficient algorithms to determine an optimal solution within a reasonable amount of time. In this course, we will focus on providing lower and upper bounds for the objective function values of such problems by studying relaxations and heuristics. On the one hand, several methods are discussed which are able to obtain dual bounds (upper bounds on the optimal value of maximization problems) and, on the other hand, we will discuss ways to find good feasible solutions that provide primal (lower) bounds. The first part of obtaining dual bounds will consider methods like branch-and-bound, cutting planes, valid inequalities, Lagrangian duality, and column-generation algorithms. The second part on finding primal bounds will present the basic ideas of (improved) local search heuristics, worst-case analysis of heuristics, and heuristics based on integer programming.	
Study Goals	* Understand the interplay between primal and dual bounds. * Apply the methods for obtaining primal and dual bounds to real-life problems to obtain good feasible solutions and provide a bound on the quality of the solution.	
Education Method	There will be 2 hours of lectures each week. Throughout the course, the students will work on real-life problems to which they can apply the discussed theory.	
Books	* Laurence A. Wolsey, Integer Programming, Wiley, 2021	
Assessment	The final grade consists of the following parts: * Final exam (70%) * Project (30%) For both the exam and project, your grade should be a 5 or higher to pass the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. F.M. de Oliveira Filho	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will deal with machine learning from the perspective of linear algebra and optimization. In particular, we will discuss how different machine learning techniques use classical concepts from linear algebra and optimization and how their performance can be improved by using more sophisticated linear algebra and optimization algorithms. We cover theoretical aspects as well as the implementation and application of these techniques. Among others, we will study + efficient matrix operations such as matrix decompositions and factorizations, + graph-related algebra, + first- and second-order optimization algorithms with their learning-oriented customization and regularization techniques, and + optimization on computational graphs (backpropagation).	
Study Goals	The student has gained an understanding of how linear algebra and optimization algorithms are used in and how these are important for the performance and efficiency of different machine learning techniques: clustering, classification and regression, neural networks, matrix decomposition/factorizations Furthermore, the student is able to + motivate and implement different matrix decomposition methods + analyze and explain matrix factorization methods + analyze graph-based data using linear algebraic tools + implement and customize first- and second- order unconstrained optimization methods + analyze and implement constrained and duality-based optimization algorithms + analyze and efficiently implement forward and backward propagation through neural networks	
Education Method	Lectures	
Assessment	Group assignments/projects and a written exam	
Co-Instructor	Dr. K.S. Postek	

WI4670	Extremal Combinatorics	6
Responsible Instructor	Dr. A. Bishnoi	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear algebra, graph theory, basic probability theory	
Course Contents	In Extremal Combinatorics we study problems of the form "How large (or small) can a parameter associated to a finite structure be under certain restrictions"? In this course we will study several problems of this kind and learn some exciting theorems from Ramsey theory, extremal set theory, etc. There will be an equal emphasis on the methods used to solve these problems. In particular, we will learn the probabilistic method, polynomial method and topological method.	
Study Goals	By the end of this course you will be able to: 1. Explain the proofs of some of the central results in extremal combinatorics. 2. Solve problems in extremal combinatorics using probabilistic, algebraic and topological methods. 3. Apply these methods to problems in other areas of mathematics and computer science.	
Education Method	Lectures	
Literature and Study Materials	Digital lecture notes will be provided on Brightspace	
Books	Extremal Combinatorics by Stasys Jukna Linear Algebra Methods in Combinatorics by László Babai and Péter Frankl A course in combinatorics by J. H. van Lint and R. M. Wilson	
Assessment	Written exam	
Permitted Materials during Tests	Pen and paper only	
Co-Instructor	Prof.dr. D.C. Gijswijt	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	A B.Sc. Programme in Electrical Engineering, Computer Science, or Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The global goal of this course is to get acquainted with the basics and applications of error correction coding techniques. Such techniques are applied in order to protect information against errors which may occur during transmission or storage. The specific techniques under consideration in the course are the ones discussed in the lecture notes, which may be updated from year to year according to recent developments. The emphasis will be on the basic trade-offs between efficiency, reliability, and complexity. Unless explicitly indicated, the proofs of the results are not part of the course contents (the interested student may consult books from the bibliography). In the end, the student should be capable of making choices for suitable error correction coding techniques in the context of information transmission and storage applications. The student has to demonstrate to have understood the aforementioned techniques and trade-offs by solving exercises in a closed-book written or oral exam. The level of these exercises is similar to the examples and exercises provided in the lecture notes.	
Education Method	Lectures; expected workload is 22 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 30 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Actual course information available on Brightspace.	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid bachelor level knowledge linear algebra and multivariate analysis. The student should also have knowledge of linear optimization and convex analysis to the level of being able to follow the text and do the exercises from the following: Linear Programming, A Concise Introduction, Thomas S. Ferguson: Available at https://www.math.ucla.edu/~tom/LP.pdf Chapters 1 and 2, along with the accompanying exercises. Convex Optimization, Stephen Boyd and Lieven Vandenbergh: Available at http://stanford.edu/~boyd/cvxbook/ Sections: 2.1, 2.2 and 3.1. Exercises (from the book): 2.1, 2.2, 2.12, 3.1, 3.3, 3.5 and 3.7	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be constrained by (differentiable) equality and inequality constraints as well as by convex cone constraints. Optimization problems like this occur naturally and commonly in science and engineering and also occur as relaxations of discrete optimization problems. Differentiability of the functions defining the problems allows for the use of multivariable calculus and linear algebra techniques to study the problems and to design and analyze efficient algorithms. This course aims to provide a concise introduction into the basics of continuous unconstrained, constrained and conic optimization.	
Study Goals	The student will be able to: - Prove results on (convex) optimization problems. - Solve the KKT conditions for basic constrained optimization problems. - Be able to formulate the Lagrange dual, and understand/prove basic results on these problems. - Give both sufficient and necessary optimality conditions for constrained continuous optimization problems. - Use a range of techniques to solve both unconstrained and constrained continuous optimization problems, and prove results on these techniques. - Formulate and recognize conic optimization problems, along with being able to construct their dual problems.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenbergh (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Remarks	This is a LNMB Mastermath course taught by Daniel Dadush.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Essentially, a BSc program in mathematics is assumed. Specifically, knowledge, understanding and ability in calculus (AM1040, AM1070), probability (AM1080), statistics (AM2080), optimization (Lagrange multipliers) and linear algebra (AM1030, AM2010) at the level of the TU Delft BSc program in Applied Mathematics. The MSc courses Linear Algebra and Optimization for Machine Learning (WI4635) and Statistical Inference (WI4455) are highly recommended as prerequisites. Furthermore basic programming skills in Python are required (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Some measure theory (AM2090) may be helpful but will not be assumed.	
Course Contents	The overarching goal in this course is to develop methods for estimating (or 'learning') an unknown function from data or make predictions of unseen function outputs. In the course we will introduce key ideas, models, algorithms and theories related to the subject of statistical learning. The detailed contents of the course may vary but will include: - linear regression and logistic regression - avoiding overfitting in high-dimensional models (shrinkage methods, LASSO) - kernel methods and Gaussian process regression - support vector machines - model selection - neural networks and deep learning - stochastic gradient descent Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	Upon successful completion of this course, the student 1. knows and understands the objective of statistical learning and the curse of dimensionality in the context of statistical learning; 2. can apply and analyse relevant models for statistical learning, including linear regression, logistic regression, sparse regression models, support vector machines and neural networks; 3. can apply and analyse ways for avoiding overfitting in the context of statistical learning; 4. can apply and analyse methods for extending, combining and selecting models, such as kernel methods and cross validation; 5. can apply and analyse computational methods and algorithms used in statistical learning, such as stochastic gradient descent.	
Education Method	Lectures, exercises, homework assignments, practice exam. Homework assignments will consist of theoretical exercises and computer assignments.	
Literature and Study Materials	The main reference will be: Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ Additional material is to be determined and will be made available during the course. Background reading / additional material: - Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ - Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). MIT press Cambridge. https://www.deeplearningbook.org/	
Assessment	assignments during the course (40%), final written exam (60%), sufficient grade (5.8) for both parts required	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Stochastics AM 2022

Introduction 1

Specialisation Stochastics

Compulsory orientation course:

WI4430 Martingales, Brownian Motion

Compulsory specialisation course:

WI4455 Statistical Inference

Compulsory specialisation courses (choose 2 out of 6 courses):

WI4465 Advanced Topics in Probability

WI4665 Advanced Topics in Statistics

WI4052 Risk Analysis

WI4230 Time series and Extreme Value Theory

MM Nonparametric Statistics

WI4615 Stochastic Calculus

Recommended specialisation courses:

WI4050 Uncertainty and Sensitivity Analysis

WI4138 Decision Theory/Expert Judgment

WI4156(TU) Game theory

WI4630 Statistical Learning

WI4640 High Dimensional Probability

WI4614 Stochastic Simulation

MM Asymptotic Statistics

MM Stochastic Processes

MM Statistics for Stochastic Processes

WI4505 Quantitative Risk Management

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4455	Statistical Inference	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Co-responsible for assignments	T.F.W. van der Jagt	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Introductory statistics and probability course at bachelor level (such as: various probability distributions, central limit theorem, estimators, maximum likelihood estimation, frequentist hypothesis testing). - Familiarity with concepts from measure theoretic probability.	
Course Contents	This course aims to provide a concise but comprehensive account of the essential elements of statistical inference and theory. The course focusses on a clear presentation of the main concepts and results underlying different frameworks of inference.	
Study Goals	Study goals Statistical Inference WI4455 The topics correspond to chapters in the lecture notes. 1. Preliminaries: - The student is able to reproduce the definition of a statistical model, possibly defined in terms of densities. - The student is able to explain the unifying concept of exponential families (the precise form of the density will be given at the exam) - The student is able to assess fundamental differences between classical- and Bayesian approaches towards problems in the given examples. - The student is able to identify with the different notions of stochastic convergence and is able to apply the Delta-method. 2. Topics from classical statistics: - The student is able to prove that a statistic is sufficient. The factorisation theorem can be proved in case of discrete random variables. - The student is able to reproduce the definition of Fisher information and Kullback-Leiber information and can prove properties of these (as in the lecture notes). - The student is able to check whether an estimator is UMVU and can prove the Lehmann-Scheffe and Cramer-Rao bound theorems. - The student can state the limiting distribution of the MLE in regular cases and can give examples of cases where the MLE is inconsistent. - The student can compute tests, p-values, power-curves of tests and errors of type I and II of (randomised) tests in simple settings. The student can prove the Neyman-Pearson lemma. - The student can state criticism to the use of p-values. - The student can derive confidence sets using (asymptotic) pivots. 3. The likelihood principle: - The student can state Birnbaums result. - The student is able to explain the implications of the sufficiency-, conditionality- and likelihood principle. 4. Bayesian analysis: - The student is able to explain the Bayesian approach towards statistical inference and can apply Bayes' rule to do Bayesian computations in case a closed form solution is available. - The student is able to describe the concept of exchangeability and its relation to justifying Bayesian analysis. - The student can derive Jeffreys prior in a given problem setting. - The student can construct hierarchical models. - The students can explain how empirical Bayes methods are defined, and can compute empirical Bayes estimators in a given problem setting. - The student can state the Bernstein-Von Mises theorem and its implications. 5. Markov Chain Monte Carlo methods: - The student can derive the steps of a MCMC scheme for drawing from the posterior, also in case a closed form solution for drawing from this posterior is unavailable. This scheme may include Gibbs steps. 6. Statistical decision theory: - The student is able to describe the basics of statistical decision theory including admissibility, minimax optimality, Bayes optimality, and connections between these topics. The student can prove results that state interrelations between these topics. - The student is able to state Steins example of inadmissibility of the MLE. - The student can derive minimax and Bayes estimators in tractable and finite-decision problems. - The student is able to derive Bayesian tests and compute a Bayes factor for a given statistical model. The exam will cover for about 25% chapters 1 and 2, the remaining 75% is on chapters 3, 4, 5 and 6	
Education Method	Each week there will be a 2-hours lecture and a question hour.	
Literature and Study Materials	Lecture notes	
Reader	Lecture notes will be available on brightspace.	
Assessment	At the end of the course there will be a written on campus exam (3 hours). The assignments in the lecture notes are representative for the level and type of questions at the exam. disclaimer: information may change depending on the developments around the coronavirus.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 6)

WI4052	Risk Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics	
Summary	Risk analysis course will treat problems often appearing in applications. Some generic methods that are applied in many different areas of applications will be discussed. We will not concentrate on any particular application, rather we will discuss important assumptions made in different models and their applicability for different real life problems. Moreover we show how parameters of these models can be obtained from data and discuss methods of validation of the obtained model.	
Course Contents	Risk modelling, life distributions, event trees, fault trees, Bayesian Belief Nets, reliability data bases, dependence modelling, software reliability, expert judgment.	
Course Contents Continuation	Introduction & Uncertainty Probability Multivariate, Copula Stochastic processes Bayesian, classical estimations & Testing Fault trees Bayesian belief nets Dependent failures Expert Judgment Reliability data bases Software reliability	
Study Goals	Students must be able to justify use of probabilistic methods in risk analysis and apply techniques like fault trees and Bayesian belief nets. They should know how to estimate parameters of models for dependent failure rates. Moreover they should be able to perform competing risks analysis for simple reliability data sets. Students should understand basic techniques of incorporating expert knowledge into risk models.	
Study Goals continuation	The goal of the course is to expose students to problems appearing in applications. During lectures the theory will be treated and exercises will be provided. Students will be required to complete the assignment where some practical problems have to be solved. The theoretical questions are not required. However making them allows to get a feedback and extra explanation in how they should be solved. These exercises together with course material will constitute basis for the exam in January. Practical exercises are required and will form part of your final grade.	
Education Method	Lectures and Assignments	
Literature and Study Materials	Parts of the book "Probabilistic Risk Analysis", Bedford & Cooke, Cam.U. press, 2001. Extra material on the Brightspace	
Reader	Brightspace	
Prerequisites	Basic probability and statistics	
Assessment	Written.	
Permitted Materials during Tests	disclaimer: information may change depending on the developments around the coronavirus. Calculator and self made A4 with formulas.	
Tags	Mathematics Stochastics	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	The final assessment will be oral or written, depending on the number of the students registered. (If there are at least 10 students we will organize a written exam.) There will be a final assessment for the extreme value theory part at the end of Q3 and a final assessment for the time series part at the end of Q4. In summer there will be the opportunity for a resit in the form of an oral exam for each part separately. There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory	

0.35 exam extreme value theory
0.15 average over assignments time series
0.35 exam time series

WI4465	Advanced Topics in Probability	6
Responsible Instructor	J. Komjáthy	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A first course in probability theory (discrete random variables, law of large numbers, central limit theorem) suffices. Knowledge of stochastic processes like Markov processes and martingales is helpful. General mathematical maturity (e.g. Bachelor in (Applied) Mathematics) is highly advised.	
Course Contents	Random graphs and complex networks. Random graphs have become one of the most studied models in probability theory. Understanding the behaviour of large networks (eg. social media, financial transactions, communication networks) has become of paramount importance for practitioners and scientists alike. In this course, we aim at providing a general theory to answer the most immediate questions about large random graphs. We will begin by recalling some tools of discrete probability (moment methods Markov's and Chebyshev's inequality), and learn some new tools: coupling and stochastic dominations. We will then study ErdosRényi graphs, one of the first examples of random networks in probability. We will then move on to more modern models of complex networks: configuration model, preferential attachment graphs, random geometric graphs.	
Course Contents Continuation	The following topics will be discussed. Note that this is only a tentative schedule and the pace at which the material will be covered will depend on the class as well. Real life complex networks and their properties, main modelling approaches. Probabilistic toolkit: Notions of convergence, coupling, stochastic ordering, Probabilistic bounds (1 lecture) Branching processes (0.5-1 lecture) Phase transition in the Erdos-Renyi random graph (1-2 lectures) Configuration model: (intro 0.5. lecture) loops and multiple edges (0.5 lecture) erased configuration model (0.5 lecture) generating simple graphs connection to number of simple graphs with prescribed degree sequence (0.5 lecture) ultra-small world property with power-law degrees. (1 lecture) Inhomogeneous random graphs (1 lectures) Preferential attachment model The model (0.5 lecture) degree distribution (1.5 lecture) Lindeberg's Central limit theorem (0.5 lecture) component sizes with Lindeberg's theorem (1 lecture) Additional topics if time allows might include small world phenomenon, etc. New models GIRGs, scale-free percolation, etc.	
Study Goals	The general goal is to make the student familiar with the probabilistic toolkit of studying random graphs and branching processes, and to get to learn the most commonly used network models. 0) The student is able to apply the following probabilistic tools in the setting of random graphs: introducing indicator variables, Markov's and Chebyshev's inequality, first and second moment method, notions of convergence of random variables, coupling and stochastic domination. 1) The student knows the definition of Galton--Watson trees and is able to derive properties of the graphs by identifying its regime (critical, subcritical, supercritical). 2) The students can recognise an Erdos--Rényi random graph and determine the size of its largest components. S/he is able to use stochastic domination and branching processes to compute expectations and variances of observables of the graph. 3) The student recognizes a configuration model, and knows efficient algorithms to generate them. They can determine and estimate basic observables: number of edges, self loops, probability of simplicity, and the relation to the number of simple graphs with given degree sequence. 4) The student knows the definition of preferential attachment models. They can apply martingale methods to determine the growth of individual degrees of vertices and give a recursion-based proof to determine the degree distribution.	
Study Goals continuation	40% of the final grade will come from a 'project'. The projects can be worked out in pairs or triplets. The possible projects are posted on bright space and will be distributed among the students on the first class. Each of the projects requires a deeper understanding and simulation of a random graph model (or a process on it). Each project has a minimal component that is required for passing grade, but beyond that, the students have the freedom to choose which direction of the project to explore. In particular, they can run simulations, or go deeper into the theory and study some papers, do calculations themselves on (partly) open questions related to random graph models. The students have to hand in a report at the end of the course, at least 5 but no longer than 12 pages. The students also present their work at the end of the course in a 10 minute presentation. (Each additional page above 12 constitutes to -0.25 grade). Projects are graded based on content and professional (writing) skills. An empty evaluation form can be found on bright space. Presentations will be (partially) peer-graded.	
Education Method	One lecture per week, to discuss the theory. We will spend every third or fourth lecture on the exercises. Exercises are provided to get familiar with the material. Beyond this, students are expected to work in pairs or triplets on their project, write a 10 page report and give a 10 min presentation (in pairs or triplets)	
Literature and Study Materials	R. van der Hofstad, "Random graphs and complex networks", volume I, available at http://www.win.tue.nl/~rhofstad/NotesRGCN.pdf	
Assessment	Written exam at the end of the semester for 60% of the grade. 40% of the grade comes from a hand-in project, executed in pairs, with a written report of max 10 pages, and a 10 min presentation.	

Co-Instructor	Prof.dr. F.H.J. Redig
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WI4615	Stochastic Calculus	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of measure theory and analysis (as given in the course ``AM2090 Real Analysis'') Martingales, Brownian motion (as taught in the master course ``WI4430 Martingales, Brownian Motion, and Stochastic Processes'').	
Course Contents	In this course the following subjects will be studied: 1. Ito integration 2. Ito calculus including Ito's formula 3. Stochastic differential equations 4. Feynman Kac formula 5. Introduction to numerical methods for stochastic differential equations	
Study Goals	After the course, you should be able to LO1. Explain the rigorous construction of the stochastic Ito integral. LO2. Demonstrate the derivation of Itos formula. LO3. Calculate explicit stochastic Ito integrals by a) using the construction of the Ito integral, b) applying Itos formula. LO4. Prove the existenceuniqueness theorem for stochastic ordinary differential equations (SODEs). LO5. Assess solvability of concrete SODEs by means of a) verifying all conditions of the existenceuniqueness theorem, b) explicitly calculating solutions. LO6. Establish links between Markov processes and partial differential equations.	
Education Method	Lectures (2 hours per week).	
Literature and Study Materials	The course will be based on the following sources: 1. P.J.C. Spreij: stochastic integration lecture notes available at https://staff.fnwi.uva.nl/p.j.c.spreij/onderwijs/master/si.pdf 2. Le Gall, Martingales, Brownian motion and stochastic calculus, Springer 2016. 3. Schilling, Partzch, Brownian Motion, An Introduction to Stochastic Processes, De Gruyter, 2010. 4. Karatzas, Shreve, Brownian Motion and Stochastic Calculus, Springer 1991. 5. Øksendal, Stochastic Differential Equations, Springer 2003. 6. Revuz, Yor, Continuous Martingales and Brownian Motion, Springer 1999.	
Assessment	These books can be freely downloaded via a library link Depending on the number of students, there will be an oral or a written exam at the end of Q4. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam is closed book.	

WI4665	Advanced Topics in Statistics	6
Responsible Instructor	Prof.dr. A.W. van der Vaart	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	The subtitle of the course is Causality and Graphical Models A graphical model expresses dependence relations between variables. Each node of a graph represents a variable and each edge or arrow a relationship. Next to this structure there is a quantitative model in the form of a joint probability distribution of all variables. The graph expresses conditional (in)dependencies between the variables given other variables in the sense of probability. Graphical models with directed graphs are also called Bayesian networks, as Bayes rule gives the conditional probabilities. They are the basis of probabilistic expert systems, as applied in many contexts. In practice a graphical model must be learned from data, guided by prior information, and hence there is a strong statistical content. The graph helps to form a likelihood, and one can apply various statistical methods, including maximum likelihood and Bayesian methods, to estimate the graph or parameters. Because there usually are many variables, statistical methods for high-dimensional data and models can be necessary. This includes for instance efficiency theory for semiparametric models and penalized likelihood methods. A main application considered in this course is the discovery and estimation of causal effects from data. Cause and effect relationships between variables can be graphically represented as arrows in a graphical model for the variables. Causal discovery is concerned with inferring such a graph from observed data, and causal estimation with quantifying the effects. The challenge is to perform this with data that is not produced by a designed experiment, but observational, a situation that abounds in epidemiological and social science research. As correlation is not causation, corrections must be made for other possible explanations that might cause a correlation. Causal reasoning can alternatively be cast in terms of "potential outcomes", the effect that would have realised had a random individual or unit received a treatment of interest. Statistical methods for estimating causal effects can be justified in terms of potential outcomes. A given causal graph allows to define the relationships of such potential outcomes to other variables, and be the basis of verifying necessary assumptions on observed data. The subject of the course is at the intersection of statistics/probability and artificial intelligence. The style will be mathematical, although only a modest amount of specialised preknowledge is required, mostly a good grasp of probability and statistical methods, and of course linear algebra and analysis. We shall provide short introductions to conditioning, asymptotic methods, Bayesian methods, Gaussian models, etc. when needed.	
Study Goals	Understand statistical graphical models and their applications, in particular in causal inference.	
Education Method	Lectures. Homework problems.	
Assessment	Written exam accounts for 70 % of the final grade. The remaining 30 % is based on the grades of the homework assignments.	
Tags	Artificial intelligence Mathematics Stochastics	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4050	Uncertainty and Sensitivity Analysis	6
Responsible Instructor	Dr. D. Kurowicka	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic Probability and Statistics (containing knowledge about joint distributions)	
Course Contents	In 2016-2017 the course have been significantly transformed. We concentrated on the most challenging part of uncertainty analysis: dependence modelling. In particular, on copula models and their applications in finance and insurance. First part of the course is concerned with the theory of copulas and their use in regression, time series analysis and factor models. The second part is designed to apply these models to financial and insurance problems.	
Study Goals	Students must be able to explain on their own examples concepts such as: uncertainty, independence, and dependence. They should understand proofs of basic properties of different dependence measures and dependence models. In particular students should be able to apply concepts as univariate and multivariate uncertainty distributions, copulas. They should be able to justify which model should be used in the particular situation and explain what are pluses and minuses of different constructions from the applied point of view. Students will apply the theory studied in the first part of the course on an individual project.	
Education Method	Lectures and Projects	
Literature and Study Materials	Study material on the Brightspace	
Assessment	Oral exam after the first part/individual projects(report, presentation) .	
disclaimer: information may change depending on the developments around the coronavirus.		

WI4138	Decision Theory/Expert Judgment	6
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Probability and Statistics	
Course Contents	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Representation of uncertainty as rational preference, subjective probability, utility, Savage's representation theorem, exchangeability, De Finetti's theorem, expert judgment, scoring rules, performance evaluation, applications.	
Study Goals	1. Can explain when and why expert judgement is needed 2. Acquire an in-depth understanding of rational decision theory and the mathematical foundations for the use of expert opinion in science. 3. Be capable of designing an expert elicitation study following the Classical Model setting 4. Be capable of conducting a real-life structured expert judgment elicitation 5. Be capable to analyze the expert data. 6. Be capable to summarize the findings in a concise and accessible report and presentation for peers.	
Education Method	Lectures, videos and online exercises (MOOC) Expected workload: 1. Attending lectures and follow the online material: 28 hours 2. Oral examination: 1 hour 3. Preparing for oral examination: 36 hours 4. Project: 70 hours 5. Oral presentation of projects: 5 h 6. Preparing for lectures, homework assignments: 20 hours 7. Reading papers: 8 hours	
Literature and Study Materials	Experts in Uncertainty, Oxford. U. Press 1991.	
Assessment	Oral examination after Q3 (40% of the final grade), Project and report (40% of the final grade), oral presentation after Q4 (10% of the final grade) and short report comparing expert judgment methods in Q4 (10%).	
disclaimer: information may change depending on the developments around the coronavirus.		

WI4156(TU)	Game theory	6
Responsible Instructor	Dr.ir. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Assessment	written exam	
	disclaimer: information may change depending on the developments around the coronavirus.	

WI4505	Quantitative Risk Management	6
Responsible Instructor	F. Barsotti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The implementation of sound and reliable quantitative risk models is of vital interest for all banks and financial institutions, both from an economic and a legal point of view. This course provides a rather comprehensive treatment of the theoretical concepts and modeling tools of quantitative risk management. At the end of the course, students will master both theory and practice, and they will be able to solve real world problems. We will cover the modeling of important risks like: market risk, operational risk and model risk. All models will be analyzed in their points of strength and, most importantly, their points of weakness, to address the main deficiencies of current practice.	
Study Goals	- To understand risk, epistemologically, statistically and in practice. - To be able to use the main risk measures available and, in case, to develop new ones. - To know the basics of EVT. - To be able to model market, credit and operational risks, using the most recent techniques. - To be able to estimate most of the models we will discuss together on real data. - To know the strengths and the weaknesses of all the discussed models, i.e. to understand the importance of model risk. - To acquire useful and practical knowledge in risk management.	
Education Method	Front lectures, as necessary supplemented by online materials including videos, case studies.	
Literature and Study Materials	McNeil, Frey and Embrechts: Quantitative Risk Management, Princeton University Press, 2015.	
Prerequisites	"Financial Mathematics" (WI4079) and "Computational Finance" (WI4154).	
Assessment	Written / Oral exam (100%). If the Covid-19 restrictions allow it, on-campus; otherwise, it will be an online examination, open book, with face-to-face remote check. disclaimer: information may change depending on the developments around the coronavirus.	
Permitted Materials during Tests	Standard non graphical calculator.	
Special Information	If you are not a mathematics student, feel free to join the course, no need to ask, but be aware that this course is highly formalized and relies on good mathematical foundation knowledge. This is a course of the master in applied mathematics!	
Remarks	A proviso: it is the firm intention of the faculty to offer this course, but (as of June 2020) it is not certain whether this will actually be possible. A second one: This information is based on the "normal" situation; Corona-prevention measures may force changes affecting the way this course is taught.	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, and the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	Mini-projects (25%); final exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course.	

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Essentially, a BSc program in mathematics is assumed. Specifically, knowledge, understanding and ability in calculus (AM1040, AM1070), probability (AM1080), statistics (AM2080), optimization (Lagrange multipliers) and linear algebra (AM1030, AM2010) at the level of the TU Delft BSc program in Applied Mathematics. The MSc courses Linear Algebra and Optimization for Machine Learning (WI4635) and Statistical Inference (WI4455) are highly recommended as prerequisites. Furthermore basic programming skills in Python are required (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Some measure theory (AM2090) may be helpful but will not be assumed.	
Course Contents	The overarching goal in this course is to develop methods for estimating (or 'learning') an unknown function from data or make predictions of unseen function outputs. In the course we will introduce key ideas, models, algorithms and theories related to the subject of statistical learning. The detailed contents of the course may vary but will include: - linear regression and logistic regression - avoiding overfitting in high-dimensional models (shrinkage methods, LASSO) - kernel methods and Gaussian process regression - support vector machines - model selection - neural networks and deep learning - stochastic gradient descent	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Upon successful completion of this course, the student 1. knows and understands the objective of statistical learning and the curse of dimensionality in the context of statistical learning; 2. can apply and analyse relevant models for statistical learning, including linear regression, logistic regression, sparse regression models, support vector machines and neural networks; 3. can apply and analyse ways for avoiding overfitting in the context of statistical learning; 4. can apply and analyse methods for extending, combining and selecting models, such as kernel methods and cross validation; 5. can apply and analyse computational methods and algorithms used in statistical learning, such as stochastic gradient descent.	
Education Method	Lectures, exercises, homework assignments, practice exam. Homework assignments will consist of theoretical exercises and computer assignments.	
Literature and Study Materials	The main reference will be: Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ Additional material is to be determined and will be made available during the course. Background reading / additional material: - Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ - Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). MIT press Cambridge. https://www.deeplearningbook.org/	
Assessment	assignments during the course (40%), final written exam (60%), sufficient grade (5.8) for both parts required	

WI4640	High-dimensional probability	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic analysis and linear algebra Introduction to Probability and Statistics Basic measure theory (as taught in the course of real analysis)	
Course Contents	High dimensional probability is the study of random vectors in $\mathbb{R}^n$ where the dimension n is large. This modern research area is important in various applications such as - Large random structures: random matrices, random networks. - Statistics and machine learning - Statistical physics - Randomized algorithms - Mixing times and other phenomena in high-dimensional Markov chains An important aspect of this theory is non-asymptotic probabilistic bounds (concentration inequalities), i.e., inequalities which are or dimension free or where the role of the dimension is explicit. The course will mainly focus on various concentration inequalities and applications. Attention will also be given to Markov semigroups and interpolation methods based on them. Concrete subjects treated in the course are - Azuma Hoeffding inequality for martingale differences - Mc Diarmid inequality, bounded difference inequality - Sub-Gaussian random variables - Sub-exponential random variables - Efron-Stein inequality and tensorization of variance - Poincare inequality and the Ornstein Uhlenbeck semigroup - Log-Sobolev inequality and the Orstein Uhlenbeck semigroup - Tensorization of entropy	
Study Goals	1. Getting acquainted with the various concentration inequalities. 2. Being able to apply concentration inequalities in various contexts of random graphs, randomized algorithms, data science. 3. Getting acquainted with estimation of suprema in the context of Gaussian processes and beyond (chaining method and Dudley's bound) 4. Getting acquainted with the applications of chaining in empirical process theory. 5. Being able to present a research paper of chapter of a book on theoretical or applied aspects of the theory provided during the course.	
Education Method	Lectures and presentations of students	
Literature and Study Materials	The basic material of the course will be based on the following two sources: 1. Vershynin: High-Dimensional Probability An Introduction with Applications in Data Science (Draft available at the homepage of the author) 2. Von Handel: Probability in High Dimension. Lecture notes available at the homepage of the author. Besides, various papers and books will be made available for the presentations of the students (these will appear on the brightspace page of the course).	
Assessment	Homework and presentation (or oral exam). The weighting is 0.4 homework+ 0.6 presentation (or oral) For the presentation the evaluation criteria are 0.4 content 0.4 clarity 0.2 presentation skills	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Mathematics of Data Science AM 2022

Introduction 1

Specialisation Mathematics of Data Science

Compulsory orientation course:

WI4635 Linear Algebra and Optimisation for Machine Learning

Compulsory specialisation course:

WI4630 Statistical Learning

Compulsory specialisation courses (choose 2 out of 4 courses):

WI4203 Applied Functional Analysis

WI4640 High Dimensional Probability

WI4455 Statistical Inference

WI4614 Stochastic Simulation

Recommended specialisation courses:

MM Machine Learning Theory

WI4207 Continuous Optimization

WI4410 Advanced Discrete Optimization

WI4645 Introduction to Quantum Information and Computing

WI4230 Time series and Extreme Value Theory

CS4230 Machine Learning 2

WI4665 Advanced Topics in Statistics

WI4620 Semidefinite Optimization

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Essentially, a BSc program in mathematics is assumed. Specifically, knowledge, understanding and ability in calculus (AM1040, AM1070), probability (AM1080), statistics (AM2080), optimization (Lagrange multipliers) and linear algebra (AM1030, AM2010) at the level of the TU Delft BSc program in Applied Mathematics. The MSc courses Linear Algebra and Optimization for Machine Learning (WI4635) and Statistical Inference (WI4455) are highly recommended as prerequisites. Furthermore basic programming skills in Python are required (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Some measure theory (AM2090) may be helpful but will not be assumed.	
Course Contents	The overarching goal in this course is to develop methods for estimating (or 'learning') an unknown function from data or make predictions of unseen function outputs. In the course we will introduce key ideas, models, algorithms and theories related to the subject of statistical learning. The detailed contents of the course may vary but will include: - linear regression and logistic regression - avoiding overfitting in high-dimensional models (shrinkage methods, LASSO) - kernel methods and Gaussian process regression - support vector machines - model selection - neural networks and deep learning - stochastic gradient descent	
Study Goals	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Upon successful completion of this course, the student 1. knows and understands the objective of statistical learning and the curse of dimensionality in the context of statistical learning; 2. can apply and analyse relevant models for statistical learning, including linear regression, logistic regression, sparse regression models, support vector machines and neural networks; 3. can apply and analyse ways for avoiding overfitting in the context of statistical learning; 4. can apply and analyse methods for extending, combining and selecting models, such as kernel methods and cross validation; 5. can apply and analyse computational methods and algorithms used in statistical learning, such as stochastic gradient descent.	
Education Method	Lectures, exercises, homework assignments, practice exam. Homework assignments will consist of theoretical exercises and computer assignments.	
Literature and Study Materials	The main reference will be: Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ Additional material is to be determined and will be made available during the course. Background reading / additional material: - Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ - Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). MIT press Cambridge. https://www.deeplearningbook.org/	
Assessment	assignments during the course (40%), final written exam (60%), sufficient grade (5.8) for both parts required	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. The e-book version will be made available free of charge via the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace.	
Permitted Materials during Tests	None	

WI4455	Statistical Inference	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Co-responsible for assignments	T.F.W. van der Jagt	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Introductory statistics and probability course at bachelor level (such as: various probability distributions, central limit theorem, estimators, maximum likelihood estimation, frequentist hypothesis testing). - Familiarity with concepts from measure theoretic probability.	
Course Contents	This course aims to provide a concise but comprehensive account of the essential elements of statistical inference and theory. The course focusses on a clear presentation of the main concepts and results underlying different frameworks of inference.	
Study Goals	Study goals Statistical Inference WI4455 The topics correspond to chapters in the lecture notes. 1. Preliminaries: - The student is able to reproduce the definition of a statistical model, possibly defined in terms of densities. - The student is able to explain the unifying concept of exponential families (the precise form of the density will be given at the exam) - The student is able to assess fundamental differences between classical- and Bayesian approaches towards problems in the given examples. - The student is able to identify with the different notions of stochastic convergence and is able to apply the Delta-method. 2. Topics from classical statistics: - The student is able to prove that a statistic is sufficient. The factorisation theorem can be proved in case of discrete random variables. - The student is able to reproduce the definition of Fisher information and Kullback-Leiber information and can prove properties of these (as in the lecture notes). - The student is able to check whether an estimator is UMVU and can prove the Lehmann-Scheffe and Cramer-Rao bound theorems. - The student can state the limiting distribution of the MLE in regular cases and can give examples of cases where the MLE is inconsistent. - The student can compute tests, p-values, power-curves of tests and errors of type I and II of (randomised) tests in simple settings. The student can prove the Neyman-Pearson lemma. - The student can state criticism to the use of p-values. - The student can derive confidence sets using (asymptotic) pivots. 3. The likelihood principle: - The student can state Birnbaums result. - The student is able to explain the implications of the sufficiency-, conditionality- and likelihood principle. 4. Bayesian analysis: - The student is able to explain the Bayesian approach towards statistical inference and can apply Bayes' rule to do Bayesian computations in case a closed form solution is available. - The student is able to describe the concept of exchangeability and its relation to justifying Bayesian analysis. - The student can derive Jeffreys prior in a given problem setting. - The student can construct hierarchical models. - The students can explain how empirical Bayes methods are defined, and can compute empirical Bayes estimators in a given problem setting. - The student can state the Bernstein-Von Mises theorem and its implications. 5. Markov Chain Monte Carlo methods: - The student can derive the steps of a MCMC scheme for drawing from the posterior, also in case a closed form solution for drawing from this posterior is unavailable. This scheme may include Gibbs steps. 6. Statistical decision theory: - The student is able to describe the basics of statistical decision theory including admissibility, minimax optimality, Bayes optimality, and connections between these topics. The student can prove results that state interrelations between these topics. - The student is able to state Steins example of inadmissibility of the MLE. - The student can derive minimax and Bayes estimators in tractable and finite-decision problems. - The student is able to derive Bayesian tests and compute a Bayes factor for a given statistical model. The exam will cover for about 25% chapters 1 and 2, the remaining 75% is on chapters 3, 4, 5 and 6	
Education Method	Each week there will be a 2-hours lecture and a question hour.	
Literature and Study Materials	Lecture notes	
Reader	Lecture notes will be available on brightspace.	
Assessment	At the end of the course there will be a written on campus exam (3 hours). The assignments in the lecture notes are representative for the level and type of questions at the exam. disclaimer: information may change depending on the developments around the coronavirus.	

WI4614	Stochastic Simulation	6
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	0/0/2/2 lecture & 0/0/2/2 projects.	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Course Contents	Stochastic simulation: algorithms and analysis. Emphasis in this course is not in the first place on doing simulations, but on understanding (the workings of) simulation techniques and how simulation can be used to provide insight into stochastic problems. Since the key in efficient simulation almost always lies with the stochastic specifics of the problem, we focus on the stochastic methods that play a role in this process. Some simulations will be done in the mini-projects, to supplement and illustrate the theory. Here, you are encouraged to "bring your own problem to work on". Subjects: introduction, general aspects of stochastic simulation; generating random objects, univariate and multivariate random variables, and stochastic processes; analysis of simulation output: how to obtain estimates for quantities of interest, as well as confidence intervals; bias and small sample issues; variance reduction, especially their stochastic background and optimization; in addition, a selection of some of the following: Quasi-Monte Carlo, discretization methods, estimating derivatives via simulation.	
Study Goals	After completing this course successfully, you will be able to 1. explain the general principles of pseudo random-number generators; 2. mention some of their important properties and how they may be assessed; 3. explain and apply the inverse transform and acceptance-rejection methods; 4. simulate, i.e., generate realizations of, random variables and random vectors; 5. do the same for some stochastic processes, among others Brownian motion and the Brownian bridge; 6. process simulation output into estimates plus standard errors/CIs; both for simple and more complex simulation schemes; 7. explain the principles of a number of variance reduction methods: control variates, antithetic variates, stratified sampling, importance sampling; 8. and apply them to problems of moderate size and complexity; 9. explain what quasi-Monte Carlo methods are and how the van der Corput and Halton sequences are generated; 10. explain some of the advantages and disadvantages of QMC methods, and what can be done about the disadvantages; 11. explain three methods of estimating parameter sensitivities or derivative estimation: finite-difference approximation, pathwise derivative estimates, and the likelihood ratio method; 12. for each of them, work out the specific details of their application for small problems; In addition: 1. you will acquire some simulation experience, first-hand, including processing and presenting the results; 2. and second-hand by sharing others experience; 3. you will practice developing a critical attitude towards presented simulation results, 4. as well as asking the right questions; 5. you will acquire some insight into Where might I use/deploy what (method or trick) from the simulation toolbox to my advantage?	
Education Method	Lectures and a bi-weekly discussion/presentation/problem solving session; homework assignments;self-study.	
Literature and Study Materials	Paul Glasserman, Monte Carlo methods in Financial Engineering, 2003, ISBN 978-0-387-00451-8. Additional reference: Stochastic Simulation by Søren Asmussen and Peter Glynn, ISBN 978-0-387-69033-9 (somewhat higher level, but also more depth).	
Prerequisites	Martingales, Brownian Motion and Stochastic processes (wi4430 or equivalent course), in particular measure theoretic probability, and basic statistics, including linear regression. Familiarity with R, Matlab, or equivalent software for simulation, is recommended.	
Assessment	Mini-projects (25%); final exam (closed book; 75%); the grade for the final must be 6 or higher to pass the course.	

WI4640	High-dimensional probability	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic analysis and linear algebra Introduction to Probability and Statistics Basic measure theory (as taught in the course of real analysis)	
Course Contents	High dimensional probability is the study of random vectors in $\mathbb{R}^n$ where the dimension n is large. This modern research area is important in various applications such as - Large random structures: random matrices, random networks. - Statistics and machine learning - Statistical physics - Randomized algorithms - Mixing times and other phenomena in high-dimensional Markov chains An important aspect of this theory is non-asymptotic probabilistic bounds (concentration inequalities), i.e., inequalities which are or dimension free or where the role of the dimension is explicit. The course will mainly focus on various concentration inequalities and applications. Attention will also be given to Markov semigroups and interpolation methods based on them. Concrete subjects treated in the course are - Azuma Hoeffding inequality for martingale differences - Mc Diarmid inequality, bounded difference inequality - Sub-Gaussian random variables - Sub-exponential random variables - Efron-Stein inequality and tensorization of variance - Poincare inequality and the Ornstein Uhlenbeck semigroup - Log-Sobolev inequality and the Orstein Uhlenbeck semigroup - Tensorization of entropy	
Study Goals	1. Getting acquainted with the various concentration inequalities. 2. Being able to apply concentration inequalities in various contexts of random graphs, randomized algorithms, data science. 3. Getting acquainted with estimation of suprema in the context of Gaussian processes and beyond (chaining method and Dudley's bound) 4. Getting acquainted with the applications of chaining in empirical process theory. 5. Being able to present a research paper of chapter of a book on theoretical or applied aspects of the theory provided during the course.	
Education Method	Lectures and presentations of students	
Literature and Study Materials	The basic material of the course will be based on the following two sources: 1. Vershynin: High-Dimensional Probability An Introduction with Applications in Data Science (Draft available at the homepage of the author) 2. Von Handel: Probability in High Dimension. Lecture notes available at the homepage of the author. Besides, various papers and books will be made available for the presentations of the students (these will appear on the brightspace page of the course).	
Assessment	Homework and presentation (or oral exam). The weighting is 0.4 homework+ 0.6 presentation (or oral) For the presentation the evaluation criteria are 0.4 content 0.4 clarity 0.2 presentation skills	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

CS4230	Machine Learning 2	5
Responsible Instructor	M. Loog	
Instructor	Dr. D.M.J. Tax	
Instructor	Dr. F.A. Oliehoek	
Instructor	Dr.ir. J.H. Krijthe	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is the more advanced and research oriented follow-up to CS4220 [Machine Learning 1]. The content of the latter is, therefore, expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning theories and techniques in detail and on an advanced level. In part, the contents of the course revolves around the research focus and interests of the lecturers teaching in the course. Possible topics : learning theory, latent variable models, online learning, Markov decision processes and abstraction, semi-supervised learning abstraction, active learning, learning curves, causal reasoning and discovery...	
Study Goals	After successfully completing the course, the student is able to apply the techniques and theories that have been covered in the course. In addition, they are able to develop learning strategies for new and previously unseen situations. Moreover, the student can provide reasoned justifications for these strategies based, for instance, on theory and/or experiment.	
Education Method	Lectures + Q&A sessions	
Assessment	Grading is based on two parts. Following the lectures -- we have about 11 of those, there is an individual assignment that will be graded pass/fail. In addition, there is a written examination that will be graded on a scale from 1 to 10. You pass the course when you both have a pass for the assignment and a passing grade for the written exam. Upon passing the course, your final grade will be the grade for the exam. Finally, note that there are two possibilities offered for the written examination, but not for the report.	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid bachelor level knowledge linear algebra and multivariate analysis. The student should also have knowledge of linear optimization and convex analysis to the level of being able to follow the text and do the exercises from the following: Linear Programming, A Concise Introduction, Thomas S. Ferguson: Available at https://www.math.ucla.edu/~tom/LP.pdf Chapters 1 and 2, along with the accompanying exercises. Convex Optimization, Stephen Boyd and Lieven Vandenberghe: Available at http://stanford.edu/~boyd/cvxbook/ Sections: 2.1, 2.2 and 3.1. Exercises (from the book): 2.1, 2.2, 2.12, 3.1, 3.3, 3.5 and 3.7	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be constrained by (differentiable) equality and inequality constraints as well as by convex cone constraints. Optimization problems like this occur naturally and commonly in science and engineering and also occur as relaxations of discrete optimization problems. Differentiability of the functions defining the problems allows for the use of multivariable calculus and linear algebra techniques to study the problems and to design and analyze efficient algorithms. This course aims to provide a concise introduction into the basics of continuous unconstrained, constrained and conic optimization.	
Study Goals	The student will be able to: - Prove results on (convex) optimization problems. - Solve the KKT conditions for basic constrained optimization problems. - Be able to formulate the Lagrange dual, and understand/prove basic results on these problems. - Give both sufficient and necessary optimality conditions for constrained continuous optimization problems. - Use a range of techniques to solve both unconstrained and constrained continuous optimization problems, and prove results on these techniques. - Formulate and recognize conic optimization problems, along with being able to construct their dual problems.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenberghe (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Remarks	This is a LNMB Mastermath course taught by Daniel Dadush.	

WI4230	Time Series & Extreme Value Theory	6
Responsible Instructor	Dr. P. Chen	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of probability and statistics at the level of the BSc Applied Mathematics courses on these topics (AM1080, AM2080).	
Course Contents	The course consists of two parts: Extreme Value Theory and Time Series. The course considers questions such as: What is the chance that the price of oil will be higher than \$120/Barrel coming week? Can we expect that UK interest rates will rise next year? How can we assess if global warming is occurring? What is the chance that the daily loss of the stock price of ABN-AMRO is larger than, say 20%, which has never occurred before? How to determine the height of a dike such that the probability of a flood, i.e., the water level exceeding the dike, in a given year is 1/10,000? What is the impact of a dramatic decline in the US stock market on DAX 30 index? In this course students will build up some theoretical basis, mainly probabilistic results within the area of time series and extreme value theory. Moreover, students will learn statistical methods for dealing with non-stationary data, multivariate data and rare events and assessing extreme risks for data from various fields such as finance, actuarial science, hydrology and environmental science. Furthermore the student is capable to perform a statistical analysis with the statistical package R. The Extreme Value Theory part considers topics such as max-domain attractions, peak over threshold approach, estimations of extreme value index, tail probability and high quantile, and tail dependence and independence. The Time Series part will cover topics such as ARMA models, spectral analysis, non-stationarity, GARCH models, state space models. Implementations of statistical methods in R will be part of the course. Examples of real applications from environmental science and finance will also be discussed.	
Study Goals	Specific learning objectives for the Extreme Value Theory part are: 1. Validate the max-domain attraction conditions, that is to show if a given distribution is in the max domain of attraction or not. 2. Compute the extreme value index of a given distribution. 3. Develop estimations for tail probabilities, high quantiles, m-year return level, using different methods. 4. Prove some asymptotic results (consistency) of the estimators. 5. Evaluate the performance of estimation methods by simulation study; and apply the estimation methods to a real data set. 6. Establish the properties of exponent measure and spectral measure. Establish the link between the bivariate extreme value distribution and these two measures. 7. Characterize the extremal dependence for bivariate distributions. 8. Develop the estimation for a bivariate tail probability (failure probability). Specific learning objectives for the Time Series part are: 1. Discern whether a given time series data set can be associated to a stationary time series, and if not, perform suitable transformations to so that the resulting data corresponds to a stationary time series. 2. Understand and determine interpret theoretical properties of AR(I)MA time series models, such as the existence and uniqueness of solutions, causality and invertibility. 3. Understand and determine descriptive statistics of time series models, such as best linear predictors, autocorrelation function, partial autocorrelation function, estimators for mean and variance, and spectral density function. 4. Have general fluency in dealing with standard manipulations in time series, such as working with linear filters, computing conditional expectations and solving recurrence relations. 5. Fit a suitable time series model to time series data and assess the quality of a time series fit. 6. Have basic knowledge of advanced time series models used in financial mathematics, such as GARCH and stochastic volatility models.	
Education Method	Lectures/exercises/computer assignment	
Books	The lectures will be mainly based on: - Time Series Analysis and Its Applications: With R Examples By Robert H. Shumway, David S. Stoffer - Time Series: Theory and Methods By Brockwell, Peter J. and Davis, Richard A - Extreme Value Theory: An Introduction By Laurens de Haan and Ana Ferreira (Chapter 1, 3, 4, 6 and 7)	
Assessment	The final assessment will be oral or written, depending on the number of the students registered. (If there are at least 10 students we will organize a written exam.) There will be a final assessment for the extreme value theory part at the end of Q3 and a final assessment for the time series part at the end of Q4. In summer there will be the opportunity for a resit in the form of an oral exam for each part separately. There will also be assignments handed out during the course. The final grade can be determined by a weighed sum, with the following weights: 0.15 average over assignments extreme value theory	

0.35 exam extreme value theory
0.15 average over assignments time series
0.35 exam time series

WI4410	Advanced Discrete Optimization	6
Responsible Instructor	Prof.dr.ir. K.I. Aardal	
Instructor	D. de Laat	
Instructor	Dr.ir. L.J.J. van Iersel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Linear optimization: basic solutions and their relation to extreme points, simplex method, duality and complementary slackness Integer optimization: basic cutting planes, branch-and-bound Computational complexity: Complexity classes P and NP and NP-complete, polynomial reductions	
Course Contents	Integer optimization: Polyhedral theory; Cutting planes; Gomory mixed-integer cuts, split cuts and their relation; Lift-and-project; IP in fixed dimension: Lattices and basis reduction; IP in fixed dimension: Lenstra's algorithm and extensions Parameterized complexity: fixed-parameter tractability, bounded search, kernelization, dynamic programming, treewidth Applications in phylogenetics: phylogenetic trees, agreement forests, SPR distance, phylogenetic networks	
Study Goals	To understand and be able to use the methods and concepts described in the course. The student should also be able to prove results similar to the ones described in the course.	
Education Method	There will be 2 hours of lectures per week. Exercises will be given that the students can do at home.	
Literature and Study Materials	The material will be made available to the students. It will consist of various scientific articles and textbook chapters.	
Assessment	Written exam	
disclaimer: information may change depending on the developments around the coronavirus.		

WI4620	Semidefinite Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Semidefinite programming is a relatively recent tool in mathematical optimization and can be seen as a vast generalization of linear programming. One can define semidefinite programming as the minimization of a linear functional over positive semidefinite matrices subject to linear constraints. Only twenty years ago it became clear that one can solve semidefinite optimization problems efficiently. Since then semidefinite optimization has become a frequently used tool in mathematics of big expressive and computational power. The main focus of this course is the application of semidefinite programming techniques to mathematical problems, such as the independent set and graph coloring problems, the max-cut problem, Nesterov's approximation algorithm, Grothendieck's inequality, quantum information theory (no prior quantum knowledge assumed), and polynomial optimization. We will also discuss the duality theory of semidefinite programming including a proof of Slater's condition and the required convex analysis.	
Study Goals	See the Brightspace page of this course.	
Education Method	The course consists of lectures and a written exam.	
Assessment	The grade is based on a written exam. The resit is also a written exam.	

WI4645	Introduction to Quantum Information and Computing	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic probability theory. Linear algebra.	
Course Contents	Quantum mechanics is not a prerequisite. We outline the mathematical framework of quantum mechanics, and give an introduction to quantum information theory, quantum computing and quantum algorithms. Since this is the orientation course for the specialization 'Mathematics of quantum technology and computation', we also pay special attention to the role in quantum theory by some of the other topics covered in this specialization: functional analysis, spectral theory, representation theory, semidefinite optimization and Lie theory.	
Study Goals	- Explain the basic postulates of quantum mechanics in terms of operators on Hilbert spaces. - Analyse self-adjoint operators using spectral theory. - Analyse single particle systems (such as qubits and harmonic oscillators) in this setting, and describe composite systems using tensor products. Analyse states on composite systems in terms of entanglement, and explain the Bell inequalities. - Analyse the time evolution on (mixed) states in the Schroedinger picture as well as in the Heisenberg picture. For qubits, explain how this gives rise to the Bloch sphere. - Analyse operations between different quantum systems in terms of completely positive maps (such as qubit gates and POVMs). Explain the Stinespring theorem and the Choi/Kraus decomposition, and explain basic no-go theorems such as the no-cloning theorem. - Analyse (the performance of) simple quantum algorithms (such as FFT). - Explain the connection between symmetries, conserved quantities and Lie theory, and analyse spin and momentum in terms of rotation and translation symmetry.	
Education Method	-2 hours of lectures per week, 14 weeks in total. -Biweekly exercises.	
Literature and Study Materials	We will use the book by Nielsen and Chuang - Quantum computation and quantum information - 10th edition. In addition, we use lecture notes that will be provided via Brightspace.	
Books	Nielsen and Chuang - Quantum computation and quantum information (10e edition). Lecture notes (provided via Brightspace)	
Assessment	A written final exam.	

WI4665	Advanced Topics in Statistics	6
Responsible Instructor	Prof.dr. A.W. van der Vaart	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	The subtitle of the course is Causality and Graphical Models A graphical model expresses dependence relations between variables. Each node of a graph represents a variable and each edge or arrow a relationship. Next to this structure there is a quantitative model in the form of a joint probability distribution of all variables. The graph expresses conditional (in)dependencies between the variables given other variables in the sense of probability. Graphical models with directed graphs are also called Bayesian networks, as Bayes rule gives the conditional probabilities. They are the basis of probabilistic expert systems, as applied in many contexts. In practice a graphical model must be learned from data, guided by prior information, and hence there is a strong statistical content. The graph helps to form a likelihood, and one can apply various statistical methods, including maximum likelihood and Bayesian methods, to estimate the graph or parameters. Because there usually are many variables, statistical methods for high-dimensional data and models can be necessary. This includes for instance efficiency theory for semiparametric models and penalized likelihood methods. A main application considered in this course is the discovery and estimation of causal effects from data. Cause and effect relationships between variables can be graphically represented as arrows in a graphical model for the variables. Causal discovery is concerned with inferring such a graph from observed data, and causal estimation with quantifying the effects. The challenge is to perform this with data that is not produced by a designed experiment, but observational, a situation that abounds in epidemiological and social science research. As correlation is not causation, corrections must be made for other possible explanations that might cause a correlation. Causal reasoning can alternatively be cast in terms of "potential outcomes", the effect that would have realised had a random individual or unit received a treatment of interest. Statistical methods for estimating causal effects can be justified in terms of potential outcomes. A given causal graph allows to define the relationships of such potential outcomes to other variables, and be the basis of verifying necessary assumptions on observed data. The subject of the course is at the intersection of statistics/probability and artificial intelligence. The style will be mathematical, although only a modest amount of specialised preknowledge is required, mostly a good grasp of probability and statistical methods, and of course linear algebra and analysis. We shall provide short introductions to conditioning, asymptotic methods, Bayesian methods, Gaussian models, etc. when needed.	
Study Goals	Understand statistical graphical models and their applications, in particular in causal inference.	
Education Method	Lectures. Homework problems.	
Assessment	Written exam accounts for 70 % of the final grade. The remaining 30 % is based on the grades of the homework assignments.	
Tags	Artificial intelligence Mathematics Stochastics	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Partial Differential Equations AM 2022

Introduction 1

Specialisation Partial Differential Equations

Compulsory orientation course:

WI4655 Perturbation and Variational Methods for Partial Differential Equations

Compulsory specialisation courses (choose 3 out of 6 courses):

WI4203 Applied Functional Analysis

WI4019 Nonlinear Differential Equations

WI4520 Nonlinear Analysis and PDEs

WI4660 Dynamical Systems and Chaos

WI4430 Martingales, Brownian Motion

WI4201 Scientific Computing

Recommended specialisation courses:

WI4006 Special Functions and Representation Theory

WI4480 Internet Seminar on Evolution Equations

WI4485 Harmonic Analysis

WI4615 Stochastic Calculus

WI4046 Spectral Theory of Linear Operators

MM Calculus of Variations

WI4204 Advanced Modeling

WI4205 Applied Finite Elements

MM Introduction to Numerical Bifurcation Analysis of ODEs and Maps

MM Mathematical Biology

WI4211 Advanced Topics in Analysis

All of the compulsory specialisation courses can also be chosen as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 3 out of 6)

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced , analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	Take home exam and oral exam. The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. The take-home exam and the oral exam count both for 50% in the final grading of the course.	
Special Information	disclaimer: information may change depending on the developments around the coronavirus. the study guide information may change depending on the developments around the coronavirus.	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5.	
Co-Instructor	disclaimer: information may change depending on the developments around the coronavirus. Prof.dr.ir. M.B. van Gijzen	

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. The e-book version will be made available free of charge via the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace.	
Permitted Materials during Tests	None	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	Two graded homeworks and a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f \geq 5$. disclaimer: information may change depending on the developments around the coronavirus. The homework grade does not count for the resit.	

WI4520	Nonlinear Analysis and Partial Differential Equations	6
Responsible Instructor	Dr. M.V. Gnann	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	required: * Applied Functional Analysis WI4203 * Perturbation and Variational Methods for Partial Differential Equations WI4655 recommended: * Spectral Theory of Linear Operators WI4046 * Dynamical Systems and Chaos * Nonlinear Differential equations WI4019	
Course Contents	In the first part of the course, we will treat the Stokes and Navier-Stokes equations and construct weak and strong solutions to the latter. This will rely on spectral methods, embedding theorems for Sobolev spaces, interpolation methods, and compactness arguments. Thus, we will prove existence and uniqueness of global weak and strong solutions to the (time-dependent) Stokes equations, existence of global weak solutions to the Navier-Stokes equations, global uniqueness for the 2D Navier-Stokes equations, and existence and uniqueness of strong solutions to the Navier-Stokes equations for short times. A discussion on the third Millennium Prize Problem rounds up the first half of the lecture. In the second part of this course we give an introduction to stability theory for nonlinear partial differential equations (PDE). Determining the stability of solutions is of central importance when analyzing PDE arising in applications, as it is typically the stable solutions that are observed in practice. This analysis is particularly challenging when the models are nonlinear, and often relies on the presence of symmetries and corresponding conservation laws, which are a common phenomenon in physical models. We will draw inspiration from linearization and energy methods for stability theory in finite-dimensional dynamical systems and show how these concepts can be adapted to the infinite-dimensional setting when studying nonlinear Hamiltonian PDE. We will apply the theory to several equations modeling wave propagation, such as the Klein-Gordon equation, the Nonlinear Schrödinger equation and the Korteweg-deVries equation.	
Study Goals	After the first part of the course the student can prove existence, (short-time) uniqueness, and regularity of solutions to equations from fluid dynamics, namely * the time-independent Stokes equations * the time-dependent Stokes equations * the Navier-Stokes equations. After the second part of the course the student can * formulate some nonlinear dispersive equations as Hamiltonian systems; * derive conservation laws from symmetries of Hamiltonian system; * characterize traveling waves as critical points of the Hamiltonian functional; * prove that (constrained) minimizers of this functional are (non)linearly stable; * analyse when saddle points of this functional are (non)linearly unstable. * apply the theory to prove (in)stability of solutions of model equations	
Education Method	Lectures	
Assessment	In each of the two quarters, a grade (Q_1/Q_2) is determined based on an oral exam (O_1/O_2), counting 80%, and solutions to exercise sheets (E_1/E_2), counting 20%. The arithmetic mean of these grades determines the final grade (F). The course is passed if the average grade is at least 6 points and if in each quarter a grade of at least 5 points is obtained. Rounding is done according to TU Delft regulations. It is possible to retake the oral exam, and if necessary the lectures and exercises in the following year, for each quarter independently until the passing criterion is fulfilled. In formulas: $Q_1 = 0.8 \cdot O_1 + 0.2 \cdot E_1$ $Q_2 = 0.8 \cdot O_2 + 0.2 \cdot E_2$ $F = (Q_1 + Q_2) / 2$ Passing criterion: $F \geq 6 \text{ and } Q_1 \geq 5 \text{ and } Q_2 \geq 5$ disclaimer: information may change depending on the developments around the coronavirus.	
Remarks	This course is offered over two quarters. It is a master elective for the specialization in partial differential equations.	
Co-Instructor	Dr. A. Geyer	

WI4660	Dynamical Systems and Chaos	6
Responsible Instructor	Dr. K. Marynets	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	analysis, ordinary differential equations, partial differential equations	
Course Contents	Topics to be covered in the course include: center manifolds, normal forms, local and global bifurcations, chaos and strange attractors, Melnikovs method, Liapunov exponents, fractional differential equations.	
Study Goals	The course aims to introduce the main features of dynamical systems as models in applied mathematics described in terms of differential equations. At the end of this course the student will be able to: - apply the center manifold theory to analyse stability of a dynamical system; - simplify dynamical systems using the normal forms' method; - analyse local and global bifurcations; - prove that the dynamical system is chaotic; - apply techniques of fractional calculus.	
Education Method	Lectures	
Books	The main book, used in the course is: S. Wiggins, Introduction to Applied Nonlinear Dynamical Systems and Chaos, Springer-Verlag 2nd Edition, Hardcover ISBN 978-0-387-00177-7; eBook ISBN 978-0-387-21749-9 For the introductory topics in fractional differential equations the following book is suggested: I. Podlubny, Fractional Differential Equations, Academic Press, Hardcover ISBN 0-12-558840-2	
Assessment	During the course two assignments will be handed out, and deadlines will be given during the course for handing in these exams. At least 70% of these exercises have to be graded as good before a student can do the oral exam. The oral exams are made on appointment. The two assignments count for 50% in the final grading of the course.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended Courses

WI4006	Special Functions and Representation Theory	6
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of Group Theory and Complex Analysis.	
Course Contents	In this course we study special functions in connection with representation theory of specific Lie groups and Lie algebras. Topics include: - Gamma and Beta integrals - hypergeometric functions - representation theory of compact matrix Lie groups and their Lie algebras - representation theory of $SU(2)$ and $SO(3)$ - Jacobi polynomials and matrix elements of $SU(2)$ - tensor products and Clebsch-Gordan coefficients	
Study Goals	The student knows the material treated during the lectures and is able to apply this to several problems. The student should be able to explain the treated material.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes will be available on Brightspace.	
Assessment	Oral exam, in combination with mandatory homework assignments. Details will be announced via Brightspace. disclaimer: information may change depending on the developments around the coronavirus.	

WI4046	Spectral Theory of Linear Operators	6
Responsible Instructor	Dr. M.P.T. Caspers	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Elements of functional analysis: Banach and Hilbert spaces, bounded linear operators, compact operators, elementary properties of the spectrum of bounded operators, spectral theorem for compact self-adjoint operators on a Hilbert space. If you have followed Applied Functional Analysis, then you have the right prerequisites. A student will benefit from a course in topology, but we do not expect this as a prerequisite. We shall use the notions: compactness, Hausdorff and topological base. We will briefly discuss this and provide material for the basics that we shall need as self-study.	
Course Contents	We will certainly treat: Single operator theory and functional calculi -Holomorphic functional calculus (also known as Dunford calculus). -Gelfand-Naimark theorem for commutative unital C^*-algebras. -Spectral theorem for bounded operators. -Unbounded operators and notions as: symmetric, self-adjoint, closed, domain, dense domain, deficiency indices, self-adjoint extensions. -Spectral theorem for unbounded self-adjoint operators. -Stone's theorem on 1-parameter groups of unitary transformations. -Fredholm operators and Fredholm index. Operator algebras -C^*-algebras -Positive operators in a C^*-algebra -States on a C^*-algebra, and the GNS-construction -Elements of von Neumann algebra theory: double commutant theorem and classification of abelian von Neumann algebras	
Study Goals	At the end of the course, students should be able to: -Explain the main notions that occur in all the topics of the course content. -The student is able to explain and analyze the proofs of the most important theorems in the course, being: the Gelfand-Naimark theorem and continuous functional calculus, the spectral theorem both for bounded and unbounded operators, the correspondence between deficiency indices and symmetric extensions of operators, Stone's theorem, the GNS-construction, the double commutant theorem and the characterization of abelian von Neumann algebras. -Apply this knowledge to solve mathematical problems using the tools of spectral theory, C^*-algebras and von Neumann algebras.	
Education Method	Classroom lectures + homework exercises.	
Literature and Study Materials	We will extensively make use of the book: John B. Conway - A course in functional analysis (Springer). You should find yourself a copy of this book to follow the course.	
Assessment	There will be a take-home mid-term exam after lecture 7. Call the grade M. There will be a final exam, which in principle is a written exam. Call the grade E. If the estimated number of students that takes the exam is below 10 or if the COVID situation does not allow for a written exam then the exam may be taken orally. Your grade will be $\text{Max}(E, 0.15 \cdot M + 0.85 \cdot E)$. If you take the retake your grade will be the grade of the retake. The retake is in principle an oral exam.	
Elective	Yes	
Tags	Analysis	

WI4204	Advanced Modeling	6
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	2e semester intensieve cursus	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The participants will benefit from a solid knowledge of advanced calculus, basic knowledge of ordinary and partial differential equations and programming experience with Matlab (or an equivalent language).	
Course Contents	This modelling course is introducing the principles and practice of mathematical modelling. An important part of the course is to recognize the essential mechanisms governing a phenomenon. These mechanisms have to be translated into mathematical equations and formulas, and included into a mathematical model. This activity requires both a good understanding of the system under consideration and good mathematical skills. Mathematical modelling is essential for understanding processes in real life, complementing expensive experiments that sometimes cannot even be carried out at all. This course will concentrate on applications in physics and life sciences and focus on models in terms of partial differential equations. The course focuses on working in a small group on a specific project. The topics for project work may differ from year to year. For the academic year 2020-2021 some emphasis will be on continuum mechanics	
Study Goals	General 1. Has knowledge of mathematical modelling tools that are used for model applications in physics and for mathematical models formulated in terms of partial differential equations. 2. The student is able to recognize the principles and practice of mathematical modelling. 3. The student is able to identify the essential mechanisms governing a physical phenomenon and is able to translate these into mathematics and can include them into a mathematical model. Specific 1. Is able to make a partial differential equation non-dimensional and can apply scaling techniques to a partial differential equation. 2. Understands conservation principles and constitutive relations. 3. Is able to linearise partial differential equations in order to perform a stability analysis 4. Is able to develop a mathematical model for a practical problem and has the knowledge and maturity to analyse this model with the available mathematical tools. 5. Has the knowledge and skills for the use and assessment of mathematical models and is able to adapt the models for a specific application. 6. Is able to pose adequate questions and has a critical yet constructive attitude towards analysing and solving complex mathematical modelling problems 7. Is able to communicate verbally and in writing about mathematical research with specialists, non-specialists and other parties involved. 8. Has sufficient communicative skills to collaborate with others and discuss mathematical modelling problems and results.	
Education Method	The first lecture summarizes the mathematical modelling tools that might be useful for the projects. Furthermore during this lecture the projects are introduced by the supervisors and the groups of 2-4 students are formed. The main part of the course is offered as a one week "intensive course", fully devoted to the project. In most cases a computer program for simulating a phenomenon from physics of the life sciences has to be developed. At the end of the fourth quarter the written report of the project has to be handed in and the final presentations are scheduled.	
Literature and Study Materials	Many of the students will have a good background in continuum mechanics. The book Continuum Modeling in the Physical Sciences, E. van Groesen and J. Molenaar, Mathematical Modeling and Computation, Vol. 13, SIAM, 2007 (ISBN 978-0-898716-25-2) can be used as background material. Copies of this e-book can be borrowed from the TU-Delft library.	
Assessment	At the first lecture home assignments are given. The final deadline for submitting the home work is the start of the intensive week. The grading will be based on the individual homework (20%) and the project (80%). The grade of the project is based on the written report of the group work and on the oral presentation about the project. The following aspects of the project work will have equal weights: -The development of the model -Mathematical depth -Report and presentation disclaimer: information may change depending on the developments around the coronavirus.	
Pop-up tekst bij intekenen	Welcome to Advanced Modelling!	

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (50% of final grade) and a written exam (50% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. - No resit option will be available for the assignments. * Exam: - The written exam must be completed with a grade greater than 5.5. - Resit option will be available for the exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4211	Advanced Topics in Analysis	6
Responsible Instructor	Dr. J.T. van Velthoven	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Fourier analysis (e.g., Fourier series and transform, Plancherel's theorem, Schwartz functions and tempered distributions). Elementary functional analysis.	
Course Contents	Phase-space analysis or time-frequency analysis is a form of local Fourier analysis. This modern field of harmonic analysis is intimately related to topics in quantum mechanics and signal analysis. The aim of the course is to present the key mathematical notions and results of time-frequency analysis. A special focus will be on Gabor expansions. Topics to be treated are: - The Heisenberg uncertainty principle - Schur orthogonality relations - Stone-von Neumann theorem - Balian-Low theorem for Gabor bases - Duality theorem for Gabor frames and Riesz sequences - Time-frequency analysis of distributions - Wiener's lemma for twisted convolution Additional topics might be covered, depending on the participants' interest.	
Study Goals	- Understanding of the main concepts and notions. - Explaining and presenting the main results and their proofs. - Applying the key results to solve selected exercises and problems.	
Education Method	Classroom lectures and homework exercises.	
Literature and Study Materials	G. Folland, Harmonic analysis in phase-space, Princeton University Press, 1989. K. Gröchenig, Foundations of time-frequency analysis, Birkhäuser, 2001.	
Assessment	Oral exam	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

WI4480	Internet Seminar on Evolution Equations	9
Responsible Instructor	Dr.ir. M.C. Veraar	
Contact Hours / Week x/x/x/x	n/a	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The internet seminar is about an advanced topic within functional analysis which is usually related to evolution equations	
Study Goals	depend on year. In principle you will learn the approach to PDEs based on semigroup theory.	
Education Method	1st phase. weekly lecture notes for self-study. You have to hand in difficult assignments on weekly basis. There are weekly meetings where the difficulties are discussed. The 2nd phase is to study advanced literature on a sub-topic in a small international group. The third phase is a workshop where the theory of the 2nd phase is presented	
Assessment	based on assignment and performance during 2nd and 3rd phase disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Dr. M.P.T. Caspers	

WI4485	Harmonic Analysis	6
Responsible Instructor	Dr.ir. M.C. Veraar	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Real Analysis, Functional Analysis, basic measure theory. Fourier Analysis can be helpful in some applications, but is not a prerequisite.	
Course Contents	This course gives an introduction into real harmonic analysis. Harmonic analysis has its origin in Fourier's work on the heat equation. Nowadays, it provides important tools in tackling a large variety of problems coming from elliptic and parabolic partial differential equation, in particular in boundary value problems. The theory has close links with operator theory and spectral theory. In this course, we will develop important concepts of real harmonic analysis, including: - covering lemmas and maximal functions - basic interpolation theory - Hardy and BMO spaces - John-Nirenberg inequality - Hilbert transform, Riesz transform and Cauchy transform - Calderón-Zygmund operators (singular integral operators) - Mikhlin multiplier theorem - Littlewood-Paley theory	
Study Goals	Global objectives: After this course students are able to - state definition and properties of fundamental objects of real harmonic analysis; - apply key techniques of harmonic analysis in a different context; - connect and apply the theory to the theory of partial differential equations. Objectives per topic: After this course students are able to - apply covering lemmas in various settings; - apply maximal operators in boundedness arguments; - apply real and complex interpolation; - state what Hardy and BMO spaces are and where they are used; - explain the importance of John-Nirenberg type arguments; - verify if a given operator is a singular integral operator; - state the most important examples of singular integral operators; - show the L^p boundedness of singular integral operators; - give a proof of Mikhlin's multiplier theorem; - apply Mikhlin's multiplier theorem to elliptic PDEs; - explain the basics of Littlewood-Paley theory.	
Education Method	Lectures and exercises/assignments.	
Literature and Study Materials	lecture notes online available on brightspace	
Books	For further reading: P. Auscher, Real Harmonic Analysis, Lecture Notes, ANU. Online available at http://eview.anu.edu.au/harmonic/index.php. L. Grafakos, Classical and modern Fourier analysis. Pearson Education, Inc., Upper Saddle River, NJ, 2004. E. M. Stein, Singular integrals and differentiability properties of functions, Princeton University Press 1970. E. M. Stein, Harmonic Analysis: Real-variables methods, orthogonality and oscillatory integrals, Princeton University Press, 1993.	
Assessment	Oral exam, in combination with homework assignments. Details will be announced via Brightspace. disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

WI4615	Stochastic Calculus	6
Responsible Instructor	Dr. K. Kirchner	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of measure theory and analysis (as given in the course ``AM2090 Real Analysis'') Martingales, Brownian motion (as taught in the master course ``W14430 Martingales, Brownian Motion, and Stochastic Processes'').	
Course Contents	In this course the following subjects will be studied: 1. Ito integration 2. Ito calculus including Ito's formula 3. Stochastic differential equations 4. Feynman Kac formula 5. Introduction to numerical methods for stochastic differential equations	
Study Goals	After the course, you should be able to LO1. Explain the rigorous construction of the stochastic Ito integral. LO2. Demonstrate the derivation of Ito's formula. LO3. Calculate explicit stochastic Ito integrals by a) using the construction of the Ito integral, b) applying Ito's formula. LO4. Prove the existenceuniqueness theorem for stochastic ordinary differential equations (SODEs). LO5. Assess solvability of concrete SODEs by means of a) verifying all conditions of the existenceuniqueness theorem, b) explicitly calculating solutions. LO6. Establish links between Markov processes and partial differential equations.	
Education Method	Lectures (2 hours per week).	
Literature and Study Materials	The course will be based on the following sources: 1. P.J.C. Spreij: stochastic integration lecture notes available at https://staff.fnwi.uva.nl/p.j.c.spreij/onderwijs/master/si.pdf 2. Le Gall, Martingales, Brownian motion and stochastic calculus, Springer 2016. 3. Schilling, Partzch, Brownian Motion, An Introduction to Stochastic Processes, De Gruyter, 2010. 4. Karatzas, Shreve, Brownian Motion and Stochastic Calculus, Springer 1991. 5. Øksendal, Stochastic Differential Equations, Springer 2003. 6. Revuz, Yor, Continuous Martingales and Brownian Motion, Springer 1999. These books can be freely downloaded via a library link	
Assessment	Depending on the number of students, there will be an oral or a written exam at the end of Q4. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	The exam is closed book.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Specialisation Mathematics of Quantum Technology and Computation AM

Introduction 1

Specialisation Mathematics of Quantum Technology and Computation

Compulsory orientation course:

WI4645 Introduction to Quantum Information and Computing

Compulsory specialisation courses (choose 2 out of 4 courses):

WI4203 Applied Functional Analysis

CS4090AM Quantum Communication and Cryptography

WI4620 Semidefinite Optimization

WI4006 Special Functions and Representation Theory

Recommended specialisation courses:

MM Quantum Computing

MM Quantum Information Theory

WI4211 Advanced Topic in Analysis

WI4046 Spectral Theory of Linear Operators

MM Operator Algebras

MM Lie Groups

WI4650 Applied Quantum Algorithms

WI4635 Linear Algebra and Optimization for Machine Learning

WI4207 Continuous Optimization

AP3663 Special Topics in Quantum Technology

All of the compulsory specialisation courses can also be chosen as as recommended specialisation course.

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Compulsory Courses (select 2 out of 4)

CS4090AM	Quantum Communication and Cryptography	6
Responsible Instructor	Dr. M. Möller	
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Introduction to Quantum Information and Computing (WI4645)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to have pre-knowledge from the course "Introduction to Quantum Information and Computing" (WI4645) given in Quarter 1 2. This year (2021/2022) the focus of this class is on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - Insight into the differences between classical and quantum communication and cryptography - Skill set valuable for other courses belonging to the Applied Mathematics master specialisation "Mathematics for Quantum Technology and Computation", e.g. WI4650 Applied Quantum Algorithms	
Education Method	Lectures and Tutorials	
Assessment	Homework (70%) and Project work (30%)	
	disclaimer: information may change depending on the developments around the coronavirus.	

WI4006	Special Functions and Representation Theory	6
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of Group Theory and Complex Analysis.	
Course Contents	In this course we study special functions in connection with representation theory of specific Lie groups and Lie algebras. Topics include: - Gamma and Beta integrals - hypergeometric functions - representation theory of compact matrix Lie groups and their Lie algebras - representation theory of SU(2) and SO(3) - Jacobi polynomials and matrix elements of SU(2) - tensor products and Clebsch-Gordan coefficients	
Study Goals	The student knows the material treated during the lectures and is able to apply this to several problems. The student should be able to explain the treated material.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes will be available on Brightspace.	
Assessment	Oral exam, in combination with mandatory homework assignments. Details will be announced via Brightspace.	
	disclaimer: information may change depending on the developments around the coronavirus.	

WI4203	Applied Functional Analysis	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The student is expected to have firm knowledge in real and complex analysis and linear algebra. Students can use the checklist below to see if he/she is well-prepared: From set theory: standard set-theoretic notions and notations, mappings and relations, injective and surjective mappings, equivalence relations and partial orderings, supremum and infimum, countable and uncountable sets. From analysis: convergence and limits, Cauchy sequences, theorem of Bolzano-Weierstrass, completeness of the fields of real and complex numbers, continuity, differentiability, convergence and absolute convergence of series, power series. From Topology: metric spaces, open and closed sets, denseness, completeness, compactness. From Integration Theory: theory of the Lebesgue integral, monotone and dominated convergence theorem, Fubini's theorem. From (linear) algebra: vector spaces over the real and complex numbers, linear mappings, subspaces, direct sum of subspaces, kernel and range of a linear mapping, linear independence, linear span, bases, dimension, polynomials, continuous functions on an interval as a vector space, inner product spaces, Cauchy-Schwarz inequality. From Complex Variables: holomorphic functions, Cauchy's integral formula and integral theorem, power series representation of holomorphic functions, analytic continuation.	
Course Contents	The course introduces those notions and results of functional analysis that are relevant for applications to, e.g., partial differential equations, numerical analysis, signal processing, probability theory, stochastic analysis, and quantum theory. More precisely, the course covers the following subjects: - Banach spaces: norms, duality, Hahn-Banach theorems. - Hilbert spaces: best approximation, orthogonality, orthocomplementation, orthonormal bases, Riesz representation theorem. - $C(K)$-spaces and L^p-spaces: completeness, duality, compactness, approximation techniques. - Bounded linear operators: uniform boundedness theorem, open mapping theorem, closed graph theorem, compact operators, adjoint operators, spectrum and resolvent, spectral projections, Riesz-Schauder theory, Fredholm operators. - Hilbert space operators: orthogonal projections, unitary operators, self-adjoint operators, normal operators, spectral theorem for compact normal operators. - weak derivatives, Sobolev spaces, applications to boundary value problems - densely defined operators, closed operators - semigroups of operators, generator, resolvent, Hille-Yosida theorem.	
Study Goals	After completing this course, the student is able to: 1) reproduce the definitions and statements of the main theorems of functional analysis as covered in the course 2) give correct and rigorous proofs of those results whose proofs do not go beyond standard reasoning from first principles 3) apply standard functional analytic proof techniques in new situations 4) name and describe some applications of functional analysis in other areas of mathematics, such as partial differential equations and quantum theory 5) communicate and present mathematical proofs clearly and understandably in a brightspace presentation	
Education Method	Lectures and Exercises/Assignments. See Brightspace for details.	
Literature and Study Materials	The course is based on my textbook "Functional Analysis", Cambridge University Press, 2022. The e-book version will be made available free of charge via the CUP website for IP addresses at the TUD. In preprint form, the book is available via my personal website.	
Assessment	Written exam, in combination with graded exercises. Details will be announced via Brightspace.	
Permitted Materials during Tests	None	

WI4620	Semidefinite Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Semidefinite programming is a relatively recent tool in mathematical optimization and can be seen as a vast generalization of linear programming. One can define semidefinite programming as the minimization of a linear functional over positive semidefinite matrices subject to linear constraints. Only twenty years ago it became clear that one can solve semidefinite optimization problems efficiently. Since then semidefinite optimization has become a frequently used tool in mathematics of big expressive and computational power. The main focus of this course is the application of semidefinite programming techniques to mathematical problems, such as the independent set and graph coloring problems, the max-cut problem, Nesterov's approximation algorithm, Grothendieck's inequality, quantum information theory (no prior quantum knowledge assumed), and polynomial optimization. We will also discuss the duality theory of semidefinite programming including a proof of Slater's condition and the required convex analysis.	
Study Goals	See the Brightspace page of this course.	
Education Method	The course consists of lectures and a written exam.	
Assessment	The grade is based on a written exam. The resit is also a written exam.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Recommended courses

AP3663	Special Topics in Quantum Technology	4
Course Coordinator	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	AP3421 Fundamentals of Quantum Information or a similar course which covers the basic model of quantum computing (qubits, quantum circuits). Optional: AP3442 Quantum Hardware I or a course which goes into the physical description of quantum computing.	
Course Contents	The content of this course changes per year and per teacher. The goal of the focus course is to provide MSc students and early PhD students more in-depth knowledge and/or tools on particular quantum hardware as pursued at QuTech. Course description for 2022/2023 "Quantum Error Correction" taught by B.M. Terhal: We discuss the ideas behind quantum error correction and fault-tolerance such as the threshold theorem, code concatenation, stabiliser and topological codes, fault-tolerant gate constructions, decoding and bosonic codes. There will some hands-on Python exercises of decoding.	
Study Goals	To get in-depth understanding of properties of qubits of a particular quantum computing platform or methods to achieve quantum computing. For 2022/2023: Obtain deep and clear understanding of the notion of quantum error correction and the tools and concept that have been developed. Understand how quantum error correction applies to real noise of various sources. Understand the idea behind fault-tolerance.	
Education Method	Weekly Lecture + Weekly Exercise Session (which includes numerical work)	
Literature and Study Materials	Lecture notes and slides.	
Assessment	60% Weekly Assignments + 40% Presentations by students	
Elective	Yes	
Tags	Physics Quantum	

WI4046	Spectral Theory of Linear Operators	6
Responsible Instructor	Dr. M.P.T. Caspers	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Elements of functional analysis: Banach and Hilbert spaces, bounded linear operators, compact operators, elementary properties of the spectrum of bounded operators, spectral theorem for compact self-adjoint operators on a Hilbert space. If you have followed Applied Functional Analysis, then you have the right prerequisites. A student will benefit from a course in topology, but we do not expect this as a prerequisite. We shall use the notions: compactness, Hausdorff and topological base. We will briefly discuss this and provide material for the basics that we shall need as self-study.	
Course Contents	We will certainly treat: Single operator theory and functional calculi -Holomorphic functional calculus (also known as Dunford calculus). -Gelfand-Naimark theorem for commutative unital C^*-algebras. -Spectral theorem for bounded operators. -Unbounded operators and notions as: symmetric, self-adjoint, closed, domain, dense domain, deficiency indices, self-adjoint extensions. -Spectral theorem for unbounded self-adjoint operators. -Stone's theorem on 1-parameter groups of unitary transformations. -Fredholm operators and Fredholm index. Operator algebras -C^*-algebras -Positive operators in a C^*-algebra -States on a C^*-algebra, and the GNS-construction -Elements of von Neumann algebra theory: double commutant theorem and classification of abelian von Neumann algebras	
Study Goals	At the end of the course, students should be able to: -Explain the main notions that occur in all the topics of the course content. -The student is able to explain and analyze the proofs of the most important theorems in the course, being: the Gelfand-Naimark theorem and continuous functional calculus, the spectral theorem both for bounded and unbounded operators, the correspondence between deficiency indices and symmetric extensions of operators, Stone's theorem, the GNS-construction, the double commutant theorem and the characterization of abelian von Neumann algebras. -Apply this knowledge to solve mathematical problems using the tools of spectral theory, C^*-algebras and von Neumann algebras.	
Education Method	Classroom lectures + homework exercises.	
Literature and Study Materials	We will extensively make use of the book: John B. Conway - A course in functional analysis (Springer). You should find yourself a copy of this book to follow the course.	
Assessment	There will be a take-home mid-term exam after lecture 7. Call the grade M. There will be a final exam, which in principle is a written exam. Call the grade E. If the estimated number of students that takes the exam is below 10 or if the COVID situation does not allow for a written exam then the exam may be taken orally. Your grade will be $\text{Max}(E, 0.15 \cdot M + 0.85 \cdot E)$. If you take the retake your grade will be the grade of the retake. The retake is in principle an oral exam.	
Elective	Yes	
Tags	Analysis	

WI4207	Continuous Optimization	6
Responsible Instructor	D. de Laat	
Contact Hours / Week x/x/x/x	1e semester Utrecht maandag	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	The student should have a solid bachelor level knowledge linear algebra and multivariate analysis. The student should also have knowledge of linear optimization and convex analysis to the level of being able to follow the text and do the exercises from the following: Linear Programming, A Concise Introduction, Thomas S. Ferguson: Available at https://www.math.ucla.edu/~tom/LP.pdf Chapters 1 and 2, along with the accompanying exercises. Convex Optimization, Stephen Boyd and Lieven Vandenberghe: Available at http://stanford.edu/~boyd/cvxbook/ Sections: 2.1, 2.2 and 3.1. Exercises (from the book): 2.1, 2.2, 2.12, 3.1, 3.3, 3.5 and 3.7	
Course Contents	Continuous optimization is the branch of optimization where we optimize a (differentiable) function over continuous (as opposed to discrete) variables. Here the variables can be constrained by (differentiable) equality and inequality constraints as well as by convex cone constraints. Optimization problems like this occur naturally and commonly in science and engineering and also occur as relaxations of discrete optimization problems. Differentiability of the functions defining the problems allows for the use of multivariable calculus and linear algebra techniques to study the problems and to design and analyze efficient algorithms. This course aims to provide a concise introduction into the basics of continuous unconstrained, constrained and conic optimization.	
Study Goals	The student will be able to: - Prove results on (convex) optimization problems. - Solve the KKT conditions for basic constrained optimization problems. - Be able to formulate the Lagrange dual, and understand/prove basic results on these problems. - Give both sufficient and necessary optimality conditions for constrained continuous optimization problems. - Use a range of techniques to solve both unconstrained and constrained continuous optimization problems, and prove results on these techniques. - Formulate and recognize conic optimization problems, along with being able to construct their dual problems.	
Education Method	There will be 14 lectures.	
Literature and Study Materials	- Convex Optimization by Stephen Boyd and Lieven Vandenberghe (http://stanford.edu/~boyd/cvxbook) - Additional notes distributed by the lecturer	
Assessment	The course will have a written exam that counts for 100% of the grade. The resit will also be a written exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Remarks	This is a LNMB Mastermath course taught by Daniel Dadush.	

WI4211	Advanced Topics in Analysis	6
Responsible Instructor	Dr. J.T. van Velthoven	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Fourier analysis (e.g., Fourier series and transform, Plancherel's theorem, Schwartz functions and tempered distributions). Elementary functional analysis.	
Course Contents	Phase-space analysis or time-frequency analysis is a form of local Fourier analysis. This modern field of harmonic analysis is intimately related to topics in quantum mechanics and signal analysis. The aim of the course is to present the key mathematical notions and results of time-frequency analysis. A special focus will be on Gabor expansions. Topics to be treated are: - The Heisenberg uncertainty principle - Schur orthogonality relations - Stone-von Neumann theorem - Balian-Low theorem for Gabor bases - Duality theorem for Gabor frames and Riesz sequences - Time-frequency analysis of distributions - Wiener's lemma for twisted convolution Additional topics might be covered, depending on the participants' interest.	
Study Goals	- Understanding of the main concepts and notions. - Explaining and presenting the main results and their proofs. - Applying the key results to solve selected exercises and problems.	
Education Method	Classroom lectures and homework exercises.	
Literature and Study Materials	G. Folland, Harmonic analysis in phase-space, Princeton University Press, 1989. K. Gröchenig, Foundations of time-frequency analysis, Birkhäuser, 2001.	
Assessment	Oral exam	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will deal with machine learning from the perspective of linear algebra and optimization. In particular, we will discuss how different machine learning techniques use classical concepts from linear algebra and optimization and how their performance can be improved by using more sophisticated linear algebra and optimization algorithms. We cover theoretical aspects as well as the implementation and application of these techniques. Among others, we will study + efficient matrix operations such as matrix decompositions and factorizations, + graph-related algebra, + first- and second-order optimization algorithms with their learning-oriented customization and regularization techniques, and + optimization on computational graphs (backpropagation).	
Study Goals	The student has gained an understanding of how linear algebra and optimization algorithms are used in and how these are important for the performance and efficiency of different machine learning techniques: clustering, classification and regression, neural networks, matrix decomposition/factorizations Furthermore, the student is able to + motivate and implement different matrix decomposition methods + analyze and explain matrix factorization methods + analyze graph-based data using linear algebraic tools + implement and customize first- and second- order unconstrained optimization methods + analyze and implement constrained and duality-based optimization algorithms + analyze and efficiently implement forward and backward propagation through neural networks	
Education Method	Lectures	
Assessment	Group assignments/projects and a written exam	
Co-Instructor	Dr. K.S. Postek	

WI4650	Applied Quantum Algorithms	6
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Good knowledge in linear algebra, numerical methods for differential equations and programming skills (Python, C/C++) are a must. Moreover, prior knowledge from the course "WI4645 Quantum Information and Computing" is highly recommended.	
Course Contents	This course introduces you to the fascinating field of quantum algorithms and their applications. We will discuss state-of-the-art quantum algorithms for solving linear systems of equations, linear and nonlinear differential equations, and continuous as well as combinatorial optimisation problems. The focus of this course is on quantum algorithms that are of practical interest which means that we will not only study the general working principles of these algorithms and their theoretical properties but also discuss the necessary pre- and post-processing steps (e.g. the encoding of system matrices into quantum-amenable format and the interpretation of probabilistic results) to integrate quantum algorithms into start-to-end quantum-assisted applications. Today's quantum computers are Noisy and Intermediate Scale Quantum (NISQ) devices that are very limited in their practical utility. This course will therefore focus on so-called variational hybrid classical-quantum algorithms (e.g. variational quantum linear solver (VQLS), quantum approximate optimization algorithm (QAOA), and quantum alternating operator ansatz (QAOA)) which combine quantum and classical computing and can be executed on real quantum hardware at least for small problem sizes. If time permits we will also be looking into the fascinating field of quantum machine learning (QML). The course will consist of two parts: - The first part will give an introduction into the subject through lectures and hands-on assignments in Q3. - In the second part (Q4) students will work in small teams on individual projects and apply the theory and methods taught in the first part of the course. As applied quantum computing is an emerging field, the content of the course might eventually change if novel promising approaches appear in recent literature or existing ones are demonstrated to be impractical.	
Study Goals	At the end of the course the student will 1. have a good overview of the current state-of-the-art in quantum algorithms and their applications; 2. be able to understand the recent literature on quantum algorithms; 3. be able to analyse the theoretical properties of quantum algorithms; 4. be able to implement quantum algorithms for practical applications; 5. be able to run quantum-assisted applications on quantum simulators and real NISQ devices; 6. be able to perform independent research on this topics, e.g. in a master project.	
Education Method	Lectures and hands-on assignments (theoretical analysis + code development) during Q3. Student team project in Q4 on topics related to the course's content and based on recent literature.	
Computer Use	A cloud-based programming environment for developing and testing quantum algorithms on multiple quantum simulator devices and real quantum hardware (if available) will be provided. It should be noted that free access to quantum hardware devices changes rapidly. We try hard to provide access to at least one (possibly more) real quantum hardware during the course.	
Literature and Study Materials	Lecture notes, relevant research papers and a list of recommended background material will be made available via Brightspace during the course.	
Assessment	Project presentation + plenary discussion of projects results	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Non-Mathematics Electives AM 2022

Introduction 1

The non-mathematics elective space may be used for

non-mathematics courses that show the role of mathematics in applications and society;
homologation, i.e. courses at bachelors level required to obtain the necessary prior knowledge for a course at masters level (for students with a bachelor degree different from (applied) mathematics);
courses of the verdiepingsdeel of SEC;
the Joint Interdisciplinary Project (TUD4040) or
an internship of 12 (WI5012) or 18 (WI5118) credits;

Additionally, a maximum of 3 credits may be spent on the following courses:

WM1135TU Advanced English for the University 3
TPM301A Spoken English for Group Work 2
TPM302A Advanced Spoken English for Group Work 2
TPM303A Writing in English for the University 2
TPM304A Advanced Writing in English for the University 2
TPM305A Writing a Master's Thesis in English 2

Please note:

Language courses may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

4. The thesis project is performed under supervision of (at least) one of the research groups at DIAM.

In case an external thesis project is mandatory (for students that do not have the internship or JIP as part of their IEP), the course code is WI5005EXT. The code for students that are free to choose between an external or internal project is WI5005. Students that are exempt from the requirement to gain experience as a mathematician in interdisciplinary work should use the code WI5005 for their thesis project in any case.

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris.	
Remarks	Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Advance in Academic Writing 1 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy will be given on Brightspace after enrolment.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Advance in Academic Writing 2 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy are given on Brightspace.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are expected to submit weekly writing assignments on a subject that is closely related to their field of study. An 85% attendance record is required. This means you may miss not more than one session (preferably none). If you fall behind with your coursework or cannot attend a minimum of six classes, you may be asked to leave the course. You will receive the course credits when all conditions are met.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

WI5012	Internship 12 EC	12
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	Self Study	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Students are required to gain experience as a mathematician in interdisciplinary work. This requirement can be satisfied in three ways: - Through an internship of 12 or 18 EC - Through a Joint Interdisciplinary Project - Through a graduation project performed in an interdisciplinary setting outside the Delft Institute of Applied Mathematics. Students that have been admitted through a bridging programme (i.e. with a bachelor degree different from (Applied) Mathematics) or that are following the SEC specialisation are exempt from this requirement, but are still allowed to include an internship in their IEP. When carrying out the internship at a company or institute, the following conditions apply: - The student takes part in the company or institute culture in general, including sharing the company-established balance between on-site and online work. - The student takes part in the the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...). You are only allowed to start if you have an approved IEP.	
Study Goals	Detailed information on projects, procedures and application is available on the Brightspace page of course WI5118. The student is aware of the professional, societal and ethical context in which mathematics is applied and is able to take this context into account while considering technological and societal problems. Besides that, the student is able to: - plan and carry out an internship project using mathematics for solving real life problems - report the results to the external supervisor - communicate and collaborate with an interdisciplinary team - provide a mathematical framework for the work carried out during the internship - reflect on their own role and experience during the internship	
Education Method	For the internship, students must do an interdisciplinary project at a company or at a non-mathematics department of a university. The main goal of the internship is to apply mathematical knowledge in a non-mathematical environment, not to gain mathematical knowledge. The duration of a project can be two (wi5012) or three months (wi5118). Students interested in an internship have to contact the responsible instructor Prof. dr. ir. M.B. van Gijzen (E-mail: M.B.vanGijzen@tudelft.nl). Before the project starts the student has to fill out an internship application form. Once the form is approved, the student can start with the project.	
Assessment	Assessment: - You need to write a report during the internship and send it to your company supervisor and the TU Delft coordinator on the final day of the internship. - You need to write your reflection note and testimonial and send it, together with the report, to the responsible instructor on the final day of your internship. - Your company supervisor needs to send the evaluation within two weeks after the end of the internship to the TU Delft coordinator. - The TU Delft coordinator needs to send the TU assessment rubric and the company evaluation form to the responsible instructor, within two weeks after receiving the evaluation from the company supervisor. - When the responsible instructor has received all documents, including the two assessment forms, your grade will be processed within a week. The responsible instructor is responsible for the final grading. The internship will be graded with a pass or a fail (not numerically).	
Remarks	disclaimer: information may change depending on the developments around the coronavirus. For more information about the internship see WI5118. The only difference between WI5012 and WI5118 is the period of the internship: Respectively 2 and 3 months	

WI5118	Internship 18 EC	18
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	Self Study	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Students are required to gain experience as a mathematician in interdisciplinary work. This requirement can be satisfied in three ways: - Through an internship of 12 or 18 EC - Through a Joint Interdisciplinary Project - Through a graduation project performed in an interdisciplinary setting outside the Delft Institute of Applied Mathematics. Students that have been admitted through a bridging programme (i.e. with a bachelor degree different from (Applied) Mathematics) or that are following the SEC specialisation are exempt from this requirement, but are still allowed to include an internship in their IEP. When carrying out the internship at a company or institute, the following conditions apply: - The student takes part in the company or institute culture in general, including sharing the company-established balance between on-site and online work. - The student takes part in the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...). You are only allowed to start if you have an approved IEP. Detailed information on projects, procedures and application is available on the Brightspace page of course WI5118.	
Study Goals	The student is aware of the professional, societal and ethical context in which mathematics is applied and is able to take this context into account while considering technological and societal problems. Besides that, the student is able to: - plan and carry out an internship project using mathematics for solving real life problems - report the results to the external supervisor - communicate and collaborate with an interdisciplinary team - provide a mathematical framework for the work carried out during the internship - reflect on their own role and experience during the internship	
Education Method	For the internship, students must do an interdisciplinary project at a company or at a non-mathematics department of a university. The main goal of the internship is to apply mathematical knowledge in a non-mathematical environment, not to gain mathematical knowledge. The duration of a project can be two (wi5012) or three months (wi5118). Students interested in an internship have to contact the responsible instructor Prof. dr. ir. M.B. van Gijzen (E-mail: M.B.vanGijzen@tudelft.nl). Before the project starts the student has to fill out an internship application form. Once the form is approved, the student can start with the project.	
Assessment	Assessment: - You need to write a report during the internship and send it to your company supervisor and the TU Delft coordinator on the final day of the internship. - You need to write your reflection note and testimonial and send it, together with the report, to the responsible instructor on the final day of your internship. - Your company supervisor needs to send the evaluation within two weeks after the end of the internship to the TU Delft coordinator. - The TU Delft coordinator needs to send the TU assessment rubric and the company evaluation form to the responsible instructor, within two weeks after receiving the evaluation from the company supervisor. - When the responsible instructor has received all documents, including the two assessment forms, your grade will be processed within a week. The responsible instructor is responsible for the final grading. The internship will be graded with a pass or a fail (not numerically). disclaimer: information may change depending on the developments around the coronavirus.	

WM0375TU	Ethics of Technological Risks	3
Module Manager	Prof.dr. S. Roeser	
Instructor	Prof.dr. S. Roeser	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk	
Education Method	Self-study, lectures and tutorials. Attending lectures and tutorials is obligatory.	
Assessment	On campus written exam In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam [There is also a 5 ECTS version of this course, i.e. WM0376TU; the only difference is that for the 5 ECTS version students also write an essay in addition to the exam.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with different types of academic texts academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as online dictionaries	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	There is no final exam. Assessment is based on the written assignments and on the in-class discussions and presentations. The six writing assignments each account for 10% of your final grade while your grade for the discussion and your grade for the presentation each accounts for 20% of your final grade.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems - can see these problems within the broader background of moral, legal and political philosophy - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1401TU	Ethics of Healthcare Technologies	3
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1401 (3ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Applied Mathematics

Thesis project

WI5005	Thesis Project	42
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	N/A	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MSc AM programs. It is an individual in-depth research with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. The thesis project may be executed within a research program at TU Delft, or in a suitable research institute or company. The nominal duration of a thesis project is 8 to 9 months (42 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. The thesis project is governed by the EEMCS Graduation Policy: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc	
Study Goals	- The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. - The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. - The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The student is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; - The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (DCO). To be able to get a DCO, the individual exam program (IEP) should have been approved by the Board of Examiners.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc/assessment disclaimer: information may change depending on the developments around the coronavirus.	
Judgement	- Quality of the work (research) (40% weight of the final grade), - Performance (30%), - Report (20%) - Presentation & defence (10%).	

WI5005EXT	Thesis Project (Extern, with additional internship optional)	42
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	N/A	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The master thesis project is the last study unit of the MSc AM programs. It is an individual in-depth research with sufficient academic content in the specific field of expertise chosen by the student and needs approval beforehand from the thesis supervisor. A thesis project with the code WI5005EXT must be executed outside DIAM, in a suitable research institute or company. The nominal duration of a thesis project is 8 to 9 months (42 ECTS). The student carries out the thesis project independently, with the guidance of one or more thesis supervisors, at least one of them from the scientific staff of Delft University of Technology, and under the responsibility of the full or associate professor who approved the IEP. With regard to the thesis project, the thesis supervisor may impose additional conditions on the IEP to be met before starting the thesis project. The thesis project is governed by the EEMCS Graduation Policy: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The student is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (DCO). To be able to get a DCO, the individual exam program (IEP) should have been approved by the Board of Examiners.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/assessment/	
Judgement	disclaimer: information may change depending on the developments around the coronavirus. - Quality of the work (research) (40% weight of the final grade), - Performance (30%), - Report (20%) - Presentation & defence (10%).	

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